Final Project Report

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Orbital Robotics and Distributed Space Systems
Politecnico di Torino

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1 Mission Context

Inspired by the Artemis programme, the present report focuses on the return phases of a lunar surface mission. This mission is composed of a Lunar Module (LM) that performs operations on the surface, and a Mothership (MS) waiting in orbit. The LM, after ascending from the surface, is inserted into a circular orbit with an altitude of $h_1 = 100$ km. From there, the crew performs an impulsive burn using thrusters to set the spacecraft in a Hohmann transfer orbit to reach the Mothership's orbit at an altitude of $h_2 = 400$ km. Given the real conditions in which the operations take place, the LM will not perfectly reach the MS, but there will be a certain position error between the two objects. The LM is then required to perform some Proximity Maneuvers in order to align itself with the MS and berth safely. In this phase, it is essential to consider the relative dynamics between the two objects: it is possible to model the system as a Chaser-Target problem, where the Chaser is the LM and the target is the MS. The Hill-Clohessy-Wiltshire (HCW) equations are a suitable mathematical model to tackle this problem. The report expands the analysis by studying the problem of perturbations due to J_2 and due to the gravitational field of the third body. Finally, the final berthing phase of the mission will be implemented through the use of a robotic arm, analysing the kinematics and dynamics of a 5-degree-of-freedom robotic manipulator mounted on the Mothership.

2 Part 1 - Orbital Transfer

Analytical computation of the minimum ΔV required to transfer the LM from its initial LLO to the Mothership's orbit, accounting for the initial phasing angle ϕ_0, waiting time estimation and Keplerian constants conservation through numerical integration of the equation of motion.

2.1 Methodology

In order to tackle this problem, we need to establish a reference framework and some initial conditions. At this first stage, we operate within the Kepler Restricted Two-Body Problem (KR2BP) framework, the two spacecraft's orbits are both circular, equatorial and coplanar. Their radii are defined as $R_i = R_{\text{Moon}} + h_i$. There is an initial phasing angle of ϕ_0 between the MS and the LM at the start of the transfer, measured in the direction of motion. The Table 1 summarizes all the initial conditions and constraints:

	Value	**Unit of Measure**
μ_{Moon}	4902.8	km^3/s^2
R_{Moon}	1737.4	km
h_{LM}	100	km
h_{MS}	400	km
ϕ_0	10	deg

Table 1: Initial conditions and constraints

Hohmann Transfer Maneuver Performing an Hohmann Transfer Maneuver means changing the orbital velocity of the circular orbit of the LM into a Hohmann transfer ellipse that, at apoapsis, allows

to reach the MS orbit and, in addition, the MS position itself. We need to calculate analytically the ΔV required to be imparted to the LM to obtain the required Hohmann ellipse.

We can calculate the orbits radii as we defined them earlier:

$$R_{\text{LM}} = R_{\text{Moon}} + h_{\text{LM}} \tag{1}$$
$$R_{\text{MS}} = R_{\text{Moon}} + h_{\text{MS}} \tag{2}$$

where R_{Moon} is the mean radius of the Moon, h_{LM} and h_{MS} are respectively the altitude of the LM and MS from the surface of the Moon.
We now can calculate the orbital velocity of LM and MS using the simplified formulas for circular orbits:

$$V_{\text{LM}} = \sqrt{\frac{\mu_{\text{Moon}}}{R_{\text{LM}}}} \tag{3}$$
$$V_{\text{MS}} = \sqrt{\frac{\mu_{\text{Moon}}}{R_{\text{MS}}}} \tag{4}$$

We can then calculate the semi-major axis length

$$a = \frac{R_{\text{LM}} + R_{\text{MS}}}{2} \tag{5}$$

and use it in the *vis-viva* equation to obtain the orbital velocity at periapsis and apoapsis of the Hohmann ellipse (i.e. the starting and ending point of the transfer maneuver).

$$V_p = \sqrt{\mu_{\text{Moon}} \left(\frac{2}{R_{\text{LM}}} - \frac{1}{a} \right)} \tag{6}$$
$$V_a = \sqrt{\mu_{\text{Moon}} \left(\frac{2}{R_{\text{MS}}} - \frac{1}{a} \right)} \tag{7}$$

The ΔV required at the start (ΔV_1) and at the end ΔV_2 of the maneuver can now be computed and then used to evalutate the minimum total ΔV required to perform the transfer

$$\Delta V_1 = V_p - V_{\text{LM}} \tag{8}$$
$$\Delta V_2 = V_{\text{MS}} - V_a \tag{9}$$
$$\Delta V = |\Delta V_1| + |\Delta V_2| \tag{10}$$

Finally, the time required for the maneuver (*time of flight*) can be found with the formula below

$$\Delta t = \frac{\tau}{2} = \frac{1}{2}\sqrt{\frac{4\pi^2 a^3}{\mu_{\text{Moon}}}} \tag{11}$$

where τ is the orbital period. The *time of flight* is half the orbital period because the rendezvous happens at the apoapsis of the transfer ellipse.

Required initial phase angle and waiting time The two spacecraft are required to meet in the MS orbit: the LM sweeps an angle of π (reaching the apoapsis of the transfer ellipse) and the MS has to be in the same angular position on its orbit. But given the fact that the MS is on a higher circular orbit, thus a slower orbit, the MS will sweep an angle lower than π and will not be in the right position required for the rendezvous during the *time of flight* of LM. In order to obtain a successful rendezvous, the LM will have to wait for the right relative angular position of the MS to start the maneuver: in this way, the two spacecraft will meet.

Using the equations and relations expressed before, we can compute the mean angular motion of both orbits:

$$n_{\mathrm{LM}} = \sqrt{\frac{\mu_{\mathrm{Moon}}}{R_{\mathrm{LM}}^3}}, \tag{12}$$

$$n_{\mathrm{MS}} = \sqrt{\frac{\mu_{\mathrm{Moon}}}{R_{\mathrm{MS}}^3}}. \tag{13}$$

The angle swept by the MS during the *time of flight* is

$$\theta_{\mathrm{MS}} = n_{\mathrm{MS}}\,\Delta t. \tag{14}$$

For a successful rendezvous to take place, the two spacecraft need to meet at the same angular position. The LM sweeps an angle $\theta_{\mathrm{LM}} = \pi$ during the maneuver (reaching the apoapsis of the transfer ellipse), while the angle swept by the MS (θ_{MS}) is lower than π. The required initial phase angle between the two vehicles for a perfect rendezvous is therefore

$$\phi_{\mathrm{req}} = \theta_{\mathrm{LM}} - \theta_{\mathrm{MS}}, \tag{15}$$

Since $n_{\mathrm{LM}} > n_{\mathrm{MS}}$, the LM moves faster than the MS. Consequently, the phase angle ϕ of the MS ahead of the LM *decreases* with time at the relative rate $n_{\mathrm{LM}} - n_{\mathrm{MS}}$. The current phase $\phi_0 = 10^\circ$ is less than the required phase $\phi_{\mathrm{req}} = 18.6118^\circ$ (as reported in Section 2.2), so the required phasing angle cannot be reached by a short wait: the LM must first complete an additional full relative orbit with respect to the MS before the required phasing recurs. The effective relative angle that must be traversed is therefore

$$\Delta\phi = 2\pi - \big(\phi_{\mathrm{req}} - \phi_0\big), \tag{16}$$

and the corresponding waiting time is

$$t_{\mathrm{wait}} = \frac{\Delta\phi}{n_{\mathrm{LM}} - n_{\mathrm{MS}}}, \tag{17}$$

where $n_{\mathrm{LM}} - n_{\mathrm{MS}}$ is the relative angular velocity between the Lunar Module and the Mothership.

Numerical propagation of the equation of motion over the full maneuver The motion of both spacecraft is governed by the two-body equations of motion

$$\ddot{\underline{r}} = -\frac{\mu_{\mathrm{Moon}}}{|\underline{r}|^3}\,\underline{r}, \tag{18}$$

which, expanded in the inertial Cartesian frame, read

$$\dot{x} = v_x, \qquad \dot{y} = v_y, \tag{19}$$

$$\dot{v}_x = -\frac{\mu_{\mathrm{Moon}}}{r^3}\,x, \qquad \dot{v}_y = -\frac{\mu_{\mathrm{Moon}}}{r^3}\,y, \tag{20}$$

where $r = \sqrt{x^2 + y^2}$. This system is integrated numerically with MATLAB's `ode45` solver, using tight tolerances (`RelTol` $= 10^{-12}$, `AbsTol` $= 10^{-14}$) to minimise integration error.

The simulation is structured in two sequential phases, consistently with the mission timeline established in Section 1.

Phase 1: Waiting. Both spacecraft start from their real initial conditions: at $t = 0$, the LM is at $(R_{\mathrm{LM}}, 0)$ with circular velocity V_{LM}, and the MS is at angle ϕ_0 on its orbit with circular velocity V_{MS}. They are propagated freely for t_{wait}. At the end of this phase the numerical phase angle between the two vehicles is verified to equal ϕ_{req} to machine precision.
During Phase 1, both orbits are circular and the positions are given exactly by

$$\underline{r}^{\mathrm{an}}_{\mathrm{LM}}(t) = R_{\mathrm{LM}} \begin{pmatrix} \cos(n_{\mathrm{LM}}\, t) \\ \sin(n_{\mathrm{LM}}\, t) \end{pmatrix}, \tag{21}$$

$$\underline{r}^{\mathrm{an}}_{\mathrm{MS}}(t) = R_{\mathrm{MS}} \begin{pmatrix} \cos(\phi_0 + n_{\mathrm{MS}}\, t) \\ \sin(\phi_0 + n_{\mathrm{MS}}\, t) \end{pmatrix}. \tag{22}$$

Impulsive burn. ΔV_1 is applied instantaneously at the end of Phase 1, in the prograde direction (tangent to the LM velocity vector). The LM speed jumps from V_{LM} to V_p.

Phase 2: Hohmann transfer. Equations for both spacecraft are propagated for Δt starting from the states obtained at the end of Phase 1. During Phase 2 the LM travels on the Hohmann ellipse. Its position is obtained from Kepler's equation

$$M = E - e \sin E, \qquad M = n_t\, \delta t, \tag{23}$$

where $\delta t = t - t_{\mathrm{wait}}$, $n_t = \sqrt{\mu_{\mathrm{Moon}}/a^3}$ is the mean motion of the transfer ellipse, and E is the eccentric anomaly solved iteratively via Newton-Raphson. The position in the perifocal frame is

$$\underline{r}_{\mathrm{peri}} = \begin{pmatrix} a(\cos E - e) \\ a\sqrt{1 - e^2}\, \sin E \end{pmatrix}, \tag{24}$$

and is then rotated into the inertial frame by the perigee angle θ_p, extracted directly from the numerical LM state at the burn epoch to avoid floating-point accumulation on the large quantity $n_{\mathrm{LM}}\, t_{\mathrm{wait}}$:

$$\underline{r}^{\mathrm{an}}_{\mathrm{LM}}(t) = \begin{pmatrix} \cos\theta_p & -\sin\theta_p \\ \sin\theta_p & \cos\theta_p \end{pmatrix} \underline{r}_{\mathrm{peri}}. \tag{25}$$

The MS continues on its circular orbit from the numerical angle at the burn epoch, $\theta_{\mathrm{MS},0}$:

$$\underline{r}^{\mathrm{an}}_{\mathrm{MS}}(t) = R_{\mathrm{MS}} \begin{pmatrix} \cos(\theta_{\mathrm{MS},0} + n_{\mathrm{MS}}\, \delta t) \\ \sin(\theta_{\mathrm{MS},0} + n_{\mathrm{MS}}\, \delta t) \end{pmatrix}. \tag{26}$$

To estimate the maneuver accuracy, the absolute position error for each spacecraft and the relative position error between them are defined as

$$\varepsilon_{\mathrm{LM}}(t) = \left\| \underline{r}^{\mathrm{num}}_{\mathrm{LM}}(t) - \underline{r}^{\mathrm{an}}_{\mathrm{LM}}(t) \right\|, \tag{27}$$

$$\varepsilon_{\mathrm{MS}}(t) = \left\| \underline{r}^{\mathrm{num}}_{\mathrm{MS}}(t) - \underline{r}^{\mathrm{an}}_{\mathrm{MS}}(t) \right\|, \tag{28}$$

$$\varepsilon_{\mathrm{rel}}(t) = \left\| (\underline{r}^{\mathrm{num}}_{\mathrm{LM}} - \underline{r}^{\mathrm{num}}_{\mathrm{MS}}) - (\underline{r}^{\mathrm{an}}_{\mathrm{LM}} - \underline{r}^{\mathrm{an}}_{\mathrm{MS}}) \right\|. \tag{29}$$

Time evolution of the Keplerian constants of motion The three first integrals of the two-body (Kepler) problem are the specific orbital energy, the specific angular momentum, and the eccentricity vector. They are exactly conserved on any unperturbed Keplerian arc, and their time evolution computed through the numerical solution is paradigmatic of the integrator's accuracy.

The three constants of motion are computed at each time step from the numerical state $(\underline{r}, \underline{\dot{r}})$ of the LM:

$$\varepsilon = \frac{v^2}{2} - \frac{\mu_{\text{Moon}}}{r}, \tag{30}$$

$$\underline{h} = \underline{r} \times \underline{\dot{r}}, \quad h = \|\underline{h}\|, \tag{31}$$

$$\underline{e} = \frac{\underline{\dot{r}} \times \underline{h}}{\mu_{\text{Moon}}} - \hat{r}, \quad e = \|\underline{e}\|. \tag{32}$$

In two dimensions, the angular momentum reduces to the scalar $h = x\dot{y} - y\dot{x}$, and the eccentricity vector components are

$$e_x = \frac{\dot{y}\, h}{\mu_{\text{Moon}}} - \frac{x}{r}, \qquad e_y = -\frac{\dot{x}\, h}{\mu_{\text{Moon}}} - \frac{y}{r}. \tag{33}$$

The analytical reference changes at the burn epoch t_{wait}, since the LM transitions from a circular orbit to the transfer ellipse:

$$\textit{Phase 1 (circular orbit):} \qquad \varepsilon_{\text{ref}} = -\frac{\mu_{\text{Moon}}}{2R_{\text{LM}}}, \qquad h_{\text{ref}} = R_{\text{LM}}\, V_{\text{LM}}, \qquad e_{\text{ref}} = 0. \tag{34}$$

$$\textit{Phase 2 (transfer ellipse):} \qquad \varepsilon_{\text{ref}} = -\frac{\mu_{\text{Moon}}}{2a}, \qquad h_{\text{ref}} = \sqrt{\mu_{\text{Moon}}\, p}, \qquad e_{\text{ref}} = e_t, \tag{35}$$

where $p = a(1 - e_t^2)$ is the semilatus rectum of the transfer ellipse and $e_t = (R_{\text{MS}} - R_{\text{LM}})/(R_{\text{MS}} + R_{\text{LM}})$ is its eccentricity. The jump in all three quantities at the burn epoch reflects the physical change of orbit produced by ΔV_1.

2.2 Results

All the analytical result obtained with MATLAB, rounded at the fourth decimal digit, are summarized in the following tables and plots.

Quantity	Value	Unit of Measure
t_{wait}	33987.9329	s

Table 4: Analytical result of the waiting time for a perfect rendezvous.

Quantity	Value	Unit of Measure
R_{LM}	1837.4	km
R_{MS}	2137.4	km
V_{LM}	1.6335	km/s
V_{MS}	1.5145	km/s
a	1987.4	km
V_p	1.6940	km/s
V_a	1.4562	km/s
ΔV_1	0.0605	km/s
ΔV_2	0.0583	km/s
ΔV	0.1188	km/s
Δt	3975.1701	s

Table 2: Analytical results of the Hohmann Transfer Maneuver.

Quantity	Value	Unit of Measure
θ_{LM}	180	deg
θ_{MS}	161.3882	deg
ϕ_{req}	18.6118	deg

Table 3: Analytical results of the required phase angle for a perfect rendezvous.

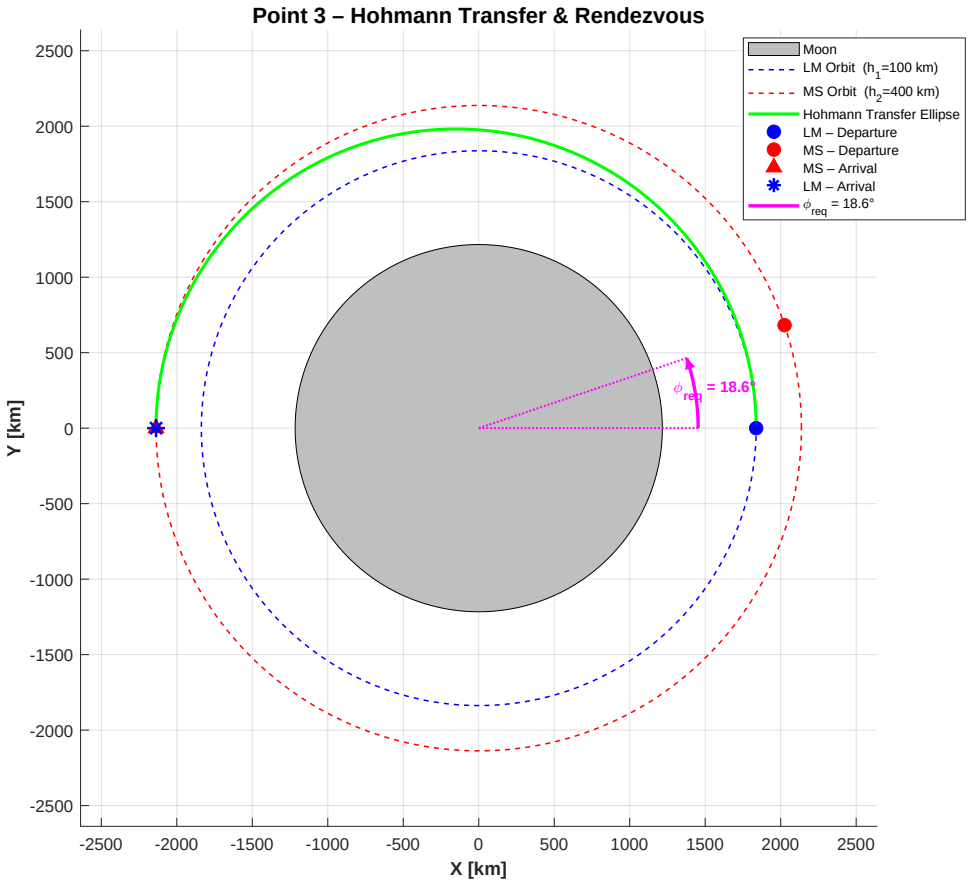


Figure 1: Visual representation of the two LLO and the Hohmann Transfer orbit

Quantity	Max value [km]	Verification
max $\varepsilon_{\mathrm{LM}}$	$\sim 10^{-10}$	PASS
max $\varepsilon_{\mathrm{MS}}$	$\sim 10^{-10}$	PASS
max $\varepsilon_{\mathrm{rel}}$	$\sim 10^{-10}$	PASS

Table 5: Maximum position errors: numerical vs. analytical solution ($\varepsilon_{\mathrm{tol}} = 10^{-6}$ km).

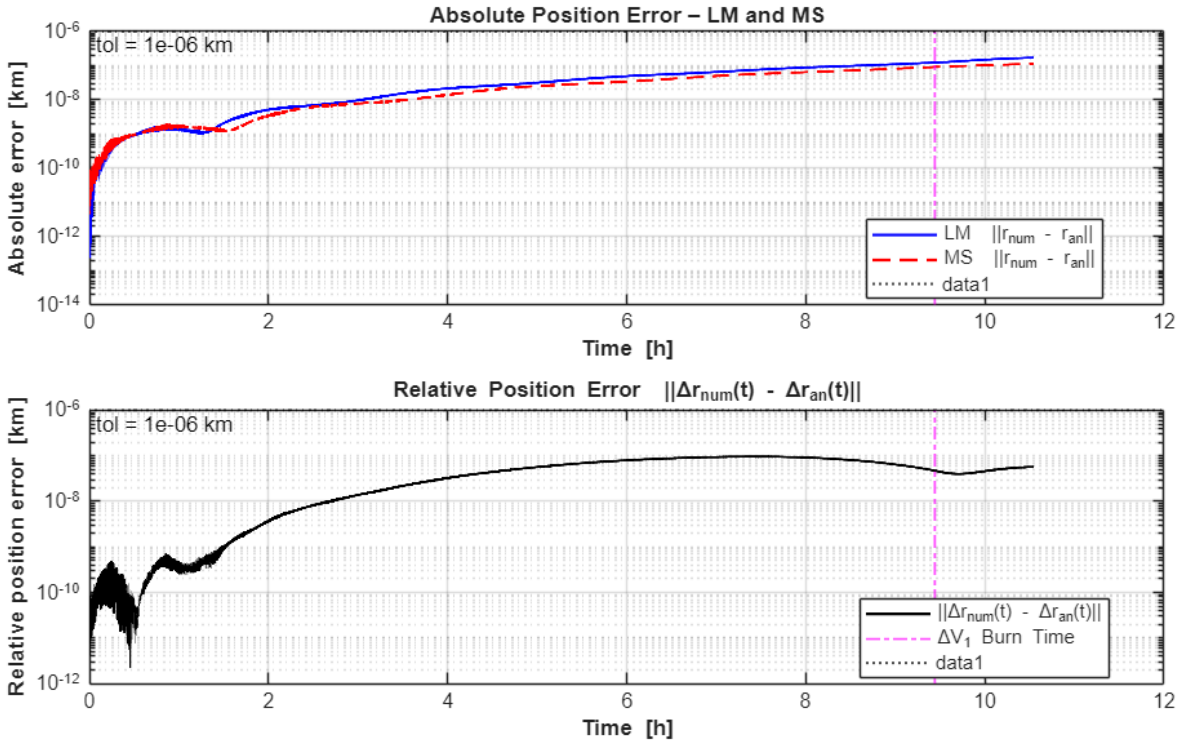


Figure 2: Absolute (top) and relative (bottom) position errors over the full maneuver. The vertical dashed line indicates the epoch of the ΔV_1 burn. All errors remain well below the 10^{-6} km tolerance.

Quantity	Value	Unit of Measure
ε	-1.33417	$\mathrm{km}^2/\mathrm{s}^2$
h	3001.40	km^2/s
e	0	-

Table 6: Reference values before the maneuver

Quantity	Value	Unit of Measure
ε	-1.23347	$\mathrm{km}^2/\mathrm{s}^2$
h	3112.61	km^2/s
e	0.0754755	-

Table 7: Reference values after the maneuver

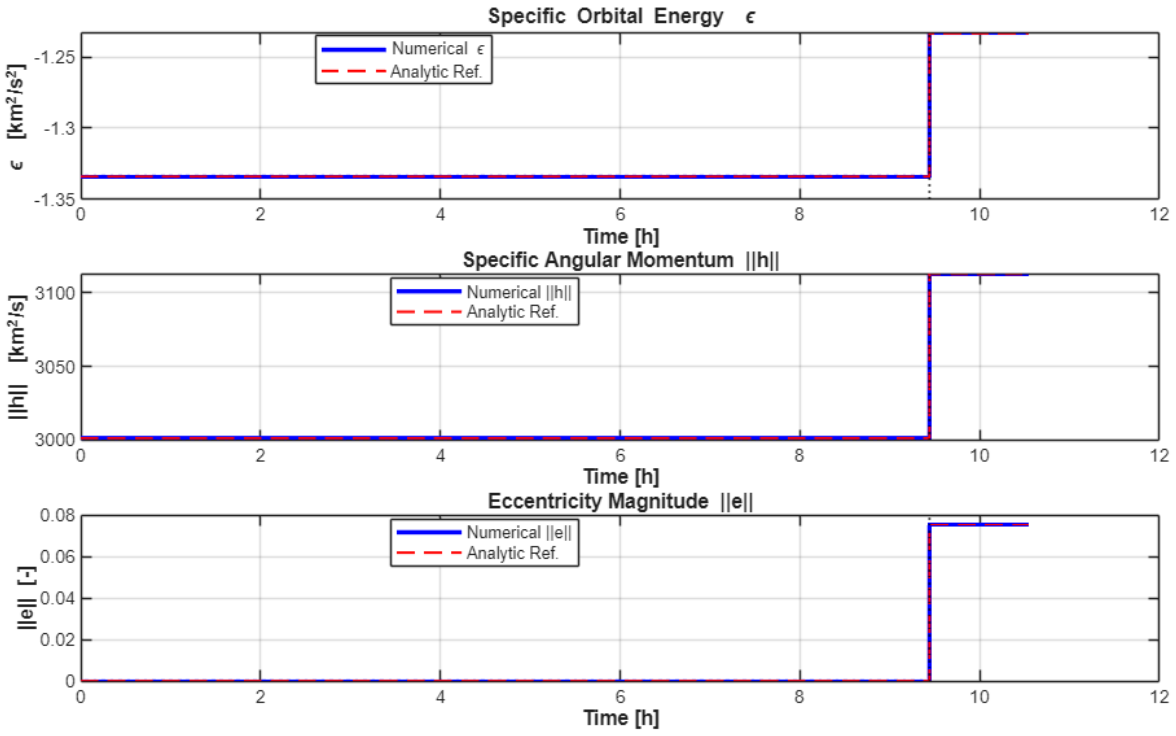


Figure 3: Time evolution of the Kepler constants of motion for the LM during the waiting phase and the Hohmann transfer. From top to bottom: specific orbital energy ε, specific angular momentum $\|\underline{h}\|$, and eccentricity magnitude $\|\underline{e}\|$. The vertical dotted line marks the impulsive maneuver ΔV_1, producing the expected discontinuous transition from the initial circular orbit to the transfer ellipse. The numerical solution (solid blue) remains constant within each orbital arc and matches the analytical reference (dashed red), confirming the accuracy of the numerical propagation.

2.3 Discussion

The obtained results demonstrate that the LM and the MS, at the beginning of the Hohmann Transfer, are not in the optimal phasing position (see Table 3) and the LM needs to wait in its orbit (see Table 4) before starting to perform the maneuver with the parameters of Table 2. The overall maneuver is plotted in Figure 1, where are evident the out-of-phase positions of the LM and MS. During the numerical propagation of the Equation of Motion, the maximum position errors of the LM, the MS and their relative distance between the numerical and the analytical solution remain below the arbitrary threshold of 10^{-10} km (Table 5). With the numerical solutions accuracy guaranteed, we can consider valid the obtained results in Table 2.

Both plots show that the errors (absolute and relative) stay well below the integration tolerance of 10^{-6} km (1 mm). The absolute error grows over time (from around 10^{-11} km to 10^{-7} km) due to the truncation and round-off errors at each integration step: this is what causes the error to accumulate. Moreover, in the first hour of the maneuver, we can recognize a noisy and oscillating behaviour, caused by the integration step adjusting at each step. Then, at the impulsive maneuver time (magenta dashed line) the relative position trend changes direction and stabilizes slightly: there is a trace of the impulsive burn in the numerical position error propagation.

During the motion, the constants of Keplerian motion change after the impulsive burn, given that the LM is now moving in an elliptical transfer orbit. Before the impulsive maneuver, the LM is on a circular LLO and the constants assume the values in Table 6. After the application of ΔV_1, the spacecraft is injected into the transfer ellipse and the constants change as reported in Table 7.

Figure 3 shows the time evolution of ε, h, and e together with their respective analytical references. Within each orbital arc, the numerical curves are flat and indistinguishable from the constant reference at the scale of the plot. The maximum numerical drift from the analytical value over the entire maneuver is reported in Table 8 and confirms that the adopted integrator preserves the Keplerian invariants to a level consistent with the chosen tolerances. The increase in specific orbital energy and angular momentum reflects the prograde impulsive burn at periapsis, while the eccentricity transitions from zero to the theoretical value of the Hohmann transfer ellipse.

Quantity	**Max drift**	**Unit**
$\lvert\varepsilon_{\text{num}} - \varepsilon_{\text{ref}}\rvert$	$\sim 10^{-12}$	km^2/s^2
$\lvert h_{\text{num}} - h_{\text{ref}}\rvert$	$\sim 10^{-9}$	km^2/s
$\lvert e_{\text{num}} - e_{\text{ref}}\rvert$	$\sim 10^{-13}$	–

Table 8: Maximum numerical drift of the Kepler constants of motion from their analytical reference values over the full maneuver.

3 Part 2 - Relative Dynamics and Proximity Operations

Use of the linearized Hill-Clohessy-Wiltshire equations to design a robust, multi-impulse proximity approach for the Lunar Module to the Mothership, validated against full non-linear models to prove the framework's accuracy in correcting residual position errors after the transfer maneuver.

3.1 Methodology

Once the transfer maneuver has been performed, the LM reaches the proximity of the MS with a relative position and velocity error. We will simulate this random error by generating a random error state vector at each code run, in order to design a truly generalized Proximity Maneuvers routine. For reproducibility reasons, the numerical values showed here correspond to a representative single run with fixed seed. In order to constrain the values generated by the MATLAB function `randn(3,1)` (that generates a 3x1 vector), we want the Euclidean norm of these values to fall within two ranges:

- Relative position error norm range: $(500 \div 1000)\,\text{m}$
- Relative velocity error norm range: $(0.1 \div 1)\,\text{m/s}$

This can be achieved by generating the random error state vector and normalizing it by its Euclidean norm; then we need to apply the range constraints to the `randn` function by creating a vector that we use for scaling. Finally, we obtain the rescaled random error state vector within the admissible range. We can note that in this way only the magnitude of the error is constrained, while the direction is unconstrained. Hence, our code needs to be general enough to tackle this problem.

We rely on the simplified framework provided by the Hill-Clohessy-Wiltshire (HCW) equations, given that we fall within its base hypothesis:

- The reference orbit is circular → the MS orbit is assumed circular by hypothesis
- The mean angular motion is constant ($|\vec{\omega}| = 0$) → the orbit radius is constant, being the orbit circular

The mean angular motion is defined as a $|\vec{\omega}| = \mathrm{n} = \sqrt{\frac{\mu}{r^3}}$.

The HCW equations expressed in the LVLH (Local Vertical Local Horizontal) frame are:

$$\begin{cases} \ddot{x} - 2n\dot{z} = f_x \\ \ddot{y} + n^2 y = f_y \\ \ddot{z} + 2n\dot{x} - 3n^2 z = f_z \end{cases} \tag{36}$$

where the in-plane motion (x-z plane) can be decoupled from the out-of-plane motion expressed by the equation in y axis. f_x, f_y, f_z are the forces imparted on each axis, in our case, by the thrusters.

In the homogeneous case ($f_x = f_y = f_z = 0$), the HCW equations have a closed form solution:

$$x(t) = \left(x_0 + \frac{2\dot{z}_0}{n}\right) - \frac{2\dot{z}_0}{n}\cos(nt) + 2\left(\frac{2\dot{x}_0}{n} - 3z_0\right)\sin(nt) + (6nz_0 - 3\dot{x}_0)t \tag{37}$$

$$y(t) = y_0\cos(nt) + \frac{\dot{y}_0}{n}\sin(nt) \tag{38}$$

$$z(t) = \left(4z_0 - \frac{2\dot{x}_0}{n}\right) + \left(\frac{2\dot{x}_0}{n} - 3z_0\right)\cos(nt) + \frac{\dot{z}_0}{n}\sin(nt) \tag{39}$$

$$\tag{40}$$

where $(x_0, y_0, z_0, \dot{x_0}, \dot{y_0}, \dot{z_0})$ are the initial conditions.

In the following sections, we will use the vector formulation of the same equations, in order to solve the problem

$$X(t) = \Phi(t)X(0) \tag{41}$$

Analytical state transition matrix We started by creating the state transfer matrix of the HCW (Hill-Clohessy-Wiltshire equations) equation from the 4 sub-matrix in order to obtain the final state vector. The relative dynamics can be written as:

$$\begin{bmatrix} \mathbf{r}(t) \\ \mathbf{v}(t) \end{bmatrix} = \begin{bmatrix} \Phi_{rr}(t) & \Phi_{rv}(t) \\ \Phi_{vr}(t) & \Phi_{vv}(t) \end{bmatrix} \begin{bmatrix} \mathbf{r}_0 \\ \mathbf{v}_0 \end{bmatrix}$$

where $\mathbf{r}_0$ is the relative initial position, $\mathbf{v}_0$ is the relative initial velocity, $\Phi_{rr}(t)$ propagates the initial position to the relative position, $\Phi_{rv}(t)$ propagates the initial velocity to the relative position, $\Phi_{vr}(t)$ propagates the initial position to the relative velocity and $\Phi_{vv}(t)$ propagates the initial velocity to the relative velocity.

The four sub-matrices describe the coupling between initial position and velocity states.

The position-to-position transition matrix is:

$$\Phi_{rr}(t) = \begin{bmatrix} 1 & 0 & 6nt - 6\sin(nt) \\ 0 & \cos(nt) & 0 \\ 0 & 0 & 4 - 3\cos(nt) \end{bmatrix}$$

The velocity-to-position transition matrix is:

$$\Phi_{rv}(t) = \begin{bmatrix} \dfrac{4\sin(nt)}{n} - 3t & 0 & \dfrac{2\,(1-\cos(nt))}{n} \\ 0 & \dfrac{\sin(nt)}{n} & 0 \\ \dfrac{2\,(\cos(nt)-1)}{n} & 0 & \dfrac{\sin(nt)}{n} \end{bmatrix}$$

The position-to-velocity transition matrix is:

$$\Phi_{vr}(t) = \begin{bmatrix} 0 & 0 & 6n\,(1-\cos(nt)) \\ 0 & -n\sin(nt) & 0 \\ 0 & 0 & 3n\sin(nt) \end{bmatrix}$$

Finally, the velocity-to-velocity transition matrix is:

$$\Phi_{vv}(t) = \begin{bmatrix} 4\cos(nt) - 3 & 0 & 2\sin(nt) \\ 0 & \cos(nt) & 0 \\ -2\sin(nt) & 0 & \cos(nt) \end{bmatrix}$$

The relative trajectory is propagated over three orbital periods of the MS using the analytical HCW State Transition Matrix.

The mean motion is computed from the circular lunar orbit:

$$n = \sqrt{\frac{\mu_{\mathrm{M}}}{r_{\mathrm{MS}}^3}}$$

where $\mu_{\mathrm{M}} = 4902.8\,\mathrm{km}^3/\mathrm{s}^2$ is the Moon gravitational parameter and $r_{\mathrm{MS}} = R_{\mathrm{M}} + h$ is the orbital radius of the MS.

Propagation of the relative motion of the LM Starting from the erroneous initial condition in the range of acceptance, the free-drift trajectory is propagated over three orbital periods of the MS using the analytical STM evaluated at each time step:

$$\delta\mathbf{x}(t_i) = \Phi(t_i)\,\delta\mathbf{x}_0. \tag{42}$$

Two-impulse corrective maneuver A two-impulse HCW maneuver is designed to transfer the LM from the erroneous post-Hohmann state ($\delta\mathbf{r}_0$, $\delta\dot{\mathbf{r}}_0$) to a V-bar hold point at 50 m ahead of the MS with zero relative velocity:

$$\mathbf{r}_f = \begin{bmatrix} 0.05 \\ 0 \\ 0 \end{bmatrix} \mathrm{km}, \qquad \dot{\mathbf{r}}_f = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \mathrm{km/s}. \tag{43}$$

The transfer time is set to $t_{man} = T/4$, one quarter of the MS orbital period. This choice ensures that $\Phi_{rv}(t_{man})$ is well-conditioned (its determinant is non-zero for $t_{man} \neq kT/2$) and provides a good balance between maneuver duration and ΔV cost.

First impulse Propagating the state through the STM with the post-burn velocity $\delta\dot{\mathbf{r}}_0 + \Delta\mathbf{v}_1$ yields

$$\mathbf{r}_f = \Phi_{rr}(t_{man})\,\delta\mathbf{r}_0 + \Phi_{rv}(t_{man})\big(\delta\dot{\mathbf{r}}_0 + \Delta\mathbf{v}_1\big). \tag{44}$$

Solving for $\Delta\mathbf{v}_1$:

$$\boxed{\Delta\mathbf{v}_1 = \Phi_{rv}^{-1}(t_{man})\big[\mathbf{r}_f - \Phi_{rr}(t_{man})\,\delta\mathbf{r}_0\big] - \delta\dot{\mathbf{r}}_0.} \tag{45}$$

Second impulse At time t_{man} the LM still has a residual relative velocity, computed via the STM:

$$\dot{\mathbf{r}}(t_{man}) = \Phi_{vr}(t_{man})\,\delta\mathbf{r}_0 + \Phi_{vv}(t_{man})\big(\delta\dot{\mathbf{r}}_0 + \Delta\mathbf{v}_1\big). \tag{46}$$

The second impulse cancels this velocity:

$$\boxed{\Delta\mathbf{v}_2 = -\dot{\mathbf{r}}(t_{man}) = -\Big[\Phi_{vr}(t_{man})\,\delta\mathbf{r}_0 + \Phi_{vv}(t_{man})\big(\delta\dot{\mathbf{r}}_0 + \Delta\mathbf{v}_1\big)\Big].} \tag{47}$$

The total ΔV needed for the two-impulse corrective maneuver is:

$$\Delta V_{tot} = \|\Delta\mathbf{v}_1\| + \|\Delta\mathbf{v}_2\|. \tag{48}$$

Final approach Starting from the hold point, the LM performs the final approach to the MS docking port (located at the LVLH origin) along the V-bar direction. The maneuver is implemented as a straight-line approach, leveraging a sequence of N_{imp} equally spaced impulsive corrections separated by Δt_{imp}.

The values adopted are $N_{imp} = 20$ and $\Delta t_{imp} = 30$ s, giving a total approach duration of $\Delta t_{tot} = N_{imp}\,\Delta t_{imp} = 600$ s. This choice is justified as follows.

- **Validity of the HCW linearisation.** The orbital angle swept by the MS between successive impulses is $n\,\Delta t_{imp} \approx 0.0213$ rad $\approx 1.1°$, small enough that the linear approximation remains accurate between corrections.
- **Approach velocity.** The mean closing speed is $\bar{V}$ is equal to $50\,\text{m}/600\,\text{s} \approx 0.083$ m/s, consistent with standard docking approach rates (< 0.1 m/s).
- **Total duration vs. orbital period.** Δt_{tot} is equal to 600 s $\approx T/15$, a small fraction of the orbital period, so HCW drift accumulated between the hold point and docking is negligible.

The implemented guidance law calculates the required impulse to obtain the desidered velocity along one direction and then the state transition matrix is updated. At each step k, the remaining transfer time is

$$t_{rem}^{(k)} = (N_{imp} - k + 1)\,\Delta t_{imp}. \tag{49}$$

A desired velocity directed purely along V-bar is imposed:

$$\mathbf{v}_{des}^{(k)} = \begin{bmatrix} \dfrac{x_{target} - x_{curr}^{(k)}}{t_{rem}^{(k)}} \\ 0 \\ 0 \end{bmatrix}. \tag{50}$$

The required impulse is

$$\Delta\mathbf{V}^{(k)} = \mathbf{v}_{des}^{(k)} - \mathbf{v}_{curr}^{(k)}. \tag{51}$$

After applying the impulse, the state is propagated over Δt_{imp} via the STM:

$$\mathbf{r}^{(k+1)} = \Phi_{rr}(\Delta t_{imp})\,\mathbf{r}^{(k)} + \Phi_{rv}(\Delta t_{imp})\,\mathbf{v}^{(k)}, \tag{52}$$

$$\mathbf{v}^{(k+1)} = \Phi_{vr}(\Delta t_{imp})\,\mathbf{r}^{(k)} + \Phi_{vv}(\Delta t_{imp})\,\mathbf{v}^{(k)}. \tag{53}$$

The guidance law enforces convergence only along the x-axis. The HCW $x - z$ coupling causes small oscillations in the radial component z, which are compensated at the next impulse step. The approach trajectory is therefore *approximately* straight along V-bar; residual lateral deviations remain at the sub-metre level throughout the maneuver, as visible in Figure 5.

The cumulative cost of the proximity maneuver is computed as:

$$\Delta V_{tot}^{(4)} = \sum_{k=1}^{N_{imp}} \left\| \Delta\mathbf{V}^{(k)} \right\|. \tag{54}$$

Verification with full non-linear equations The entire approach maneuver calculation is performed again within the full non-linear equations framework.

The HCW equations are obtained by linearising the exact non-linear relative motion equation about a circular reference orbit. Starting from the inertial relative acceleration in the LVLH rotating frame:

$$\ddot{\boldsymbol{\rho}} + 2\boldsymbol{\omega} \times \dot{\boldsymbol{\rho}} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{\rho}) = -\frac{\mu}{|\mathbf{r} + \boldsymbol{\rho}|^3}(\mathbf{r} + \boldsymbol{\rho}) + \frac{\mu}{r^3}\mathbf{r} + \mathbf{f}, \tag{55}$$

where $\mathbf{r}$ is the target (MS) position vector and $\boldsymbol{\rho}$ is the chaser (LM) relative position. For a circular reference orbit $\dot{\omega} = 0$ and $r = \text{const}$, so $\boldsymbol{\omega} = n\hat{j}$ (out-of-plane unit vector of LVLH). Linearising equation (55) for $|\boldsymbol{\rho}| \ll r$ yields the HCW equations given above. Their state-space form is $\dot{\mathbf{X}} = A\mathbf{X}$, whose matrix exponential e^{At} gives the STM $\Phi(t)$ already reported.

The full non-linear relative motion is propagated over three orbital periods using `ode45` with `RelTol` $= 10^{-12}$ and `AbsTol` $= 10^{-14}$. Note that all the output results are from a single run. The non-linear ODE used for the propagation is

$$\ddot{\boldsymbol{\rho}} = -\frac{\mu_M}{|\mathbf{r}_{MS} + \boldsymbol{\rho}|^3}(\mathbf{r}_{MS} + \boldsymbol{\rho}) + \frac{\mu_M}{r_{MS}^3}\,\mathbf{r}_{MS} - 2\boldsymbol{\omega} \times \dot{\boldsymbol{\rho}} - \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{\rho}), \tag{56}$$

with $\boldsymbol{\omega} = [0,\, -n,\, 0]^\top$ rad/s in the LVLH convention adopted ($\hat{k} = -\mathbf{r}/r$ toward the Moon, $\hat{j} = -\mathbf{h}/h$, so $\boldsymbol{\omega} = -n\hat{j}$).

The position error between the two frameworks, the linear HCW and the non-linear ODE is

$$\varepsilon(t) = \|\boldsymbol{\rho}_{NL}(t) - \boldsymbol{\rho}_{HCW}(t)\|. \tag{57}$$

The first impulse $\Delta\mathbf{v}_1^{NL}$ is recomputed by solving the non-linear targeting problem with MATLAB's `fsolve` function, using the HCW solution (45) as initial guess:

$$\min_{\Delta\mathbf{v}_1} \left\| \boldsymbol{\rho}_{NL}(t_{man};\, \delta\mathbf{r}_0,\, \delta\dot{\mathbf{r}}_0 + \Delta\mathbf{v}_1) - \mathbf{r}_f \right\|. \tag{58}$$

The second impulse cancels out the residual velocity at arrival: $\Delta\mathbf{v}_2^{NL} = -\dot{\boldsymbol{\rho}}_{NL}(t_{man})$.

The difference between the HCW and non-linear impulses quantifies the error introduced with the linearisation.

The V-bar straight-line approach is repeated propagating each segment with the full non-linear ODE (56) instead of the STM. The same guidance law is applied. The trajectory deviation between the two models is

$$\varepsilon_k = \|\mathbf{r}_k^{NL} - \mathbf{r}_k^{HCW}\|, \quad k = 1, \ldots, N_{imp}. \tag{59}$$

It is important to note that the HCW equations provide an accurate description of the relative dynamics when two conditions hold simultaneously: the reference orbit is circular, and the separation $|\boldsymbol{\rho}|$ is small compared to the orbital radius r. In this mission scenario:

- The MS orbit is circular by assumption, satisfying the first condition exactly.
- The maximum relative distance is on the order of 1 km $\ll r_{MS} \approx 2137$ km, so the ratio $|\boldsymbol{\rho}|/r \lesssim 5 \times 10^{-4}$, which keeps higher-order terms in the Taylor expansion of the gravity gradient negligible.

The numerical comparison (Point 5.2) confirms that the HCW and non-linear trajectories deviate by at most a few metres over three orbital periods, well within the acceptable error budget for the proximity operations phase. The linearisation error is therefore not operationally significant for the distances considered in this mission.

3.2 Results

Since the initial errors are generated randomly, the numerical values below correspond to a representative single run (seed fixed for reproducibility).

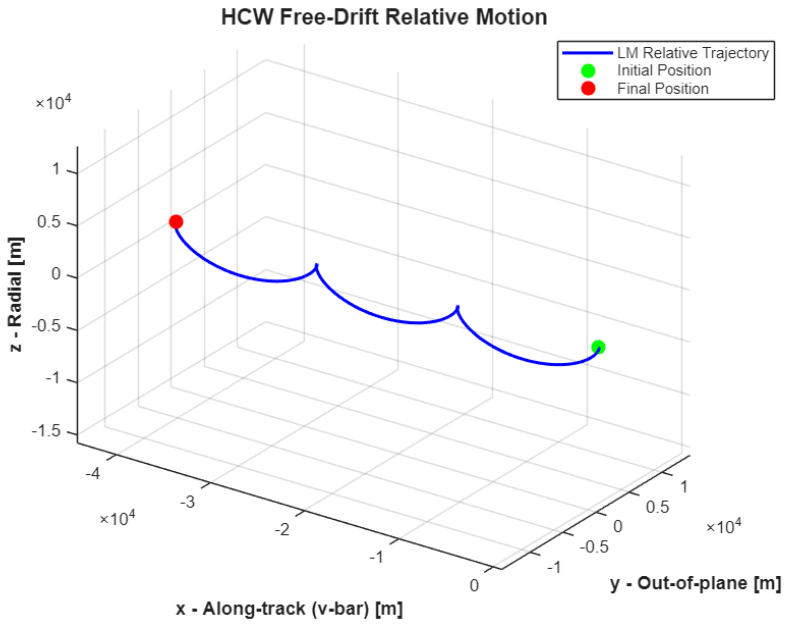


Figure 4: Free-drift relative trajectory of the Lunar Module in the LVLH frame (HCW propagation over $3T$). Green marker: initial position; red marker: final position after three orbital periods.

Quantity	Value	Unit
$\|\Delta\mathbf{v}_1\|$	1.2116	m/s
$\|\Delta\mathbf{v}_2\|$	0.3895	m/s
ΔV_{tot}	1.6012	m/s
t_{man}	$T/4 \approx 2217$	s

Table 9: Two-impulse corrective maneuver results , obtained with fixed random seed `rng(42)` for reproducibility.

Quantity	Value	Unit
N_{imp}	20	–
Δt_{imp}	30	s
Δt_{tot}	600	s
$\Delta V_{tot}^{(4)}$	0.151090	m/s

Table 10: V-bar approach parameters and total ΔV cost. The maneuver starts from the fixed hold point $[50, 0, 0]$ m, so the cost is deterministic and independent of the random initial errors.

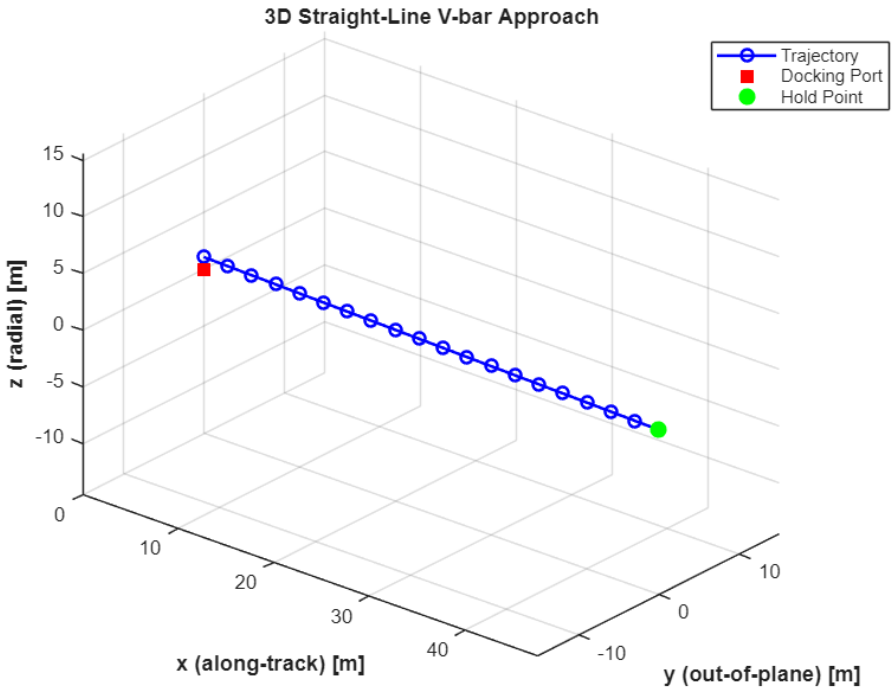


Figure 5: Three-dimensional V-bar approach trajectory in the LVLH frame. Green marker: hold point ($x = 50$ m); red marker: docking port (origin). Small radial deviations due to HCW coupling are visible and are corrected at each impulse step.

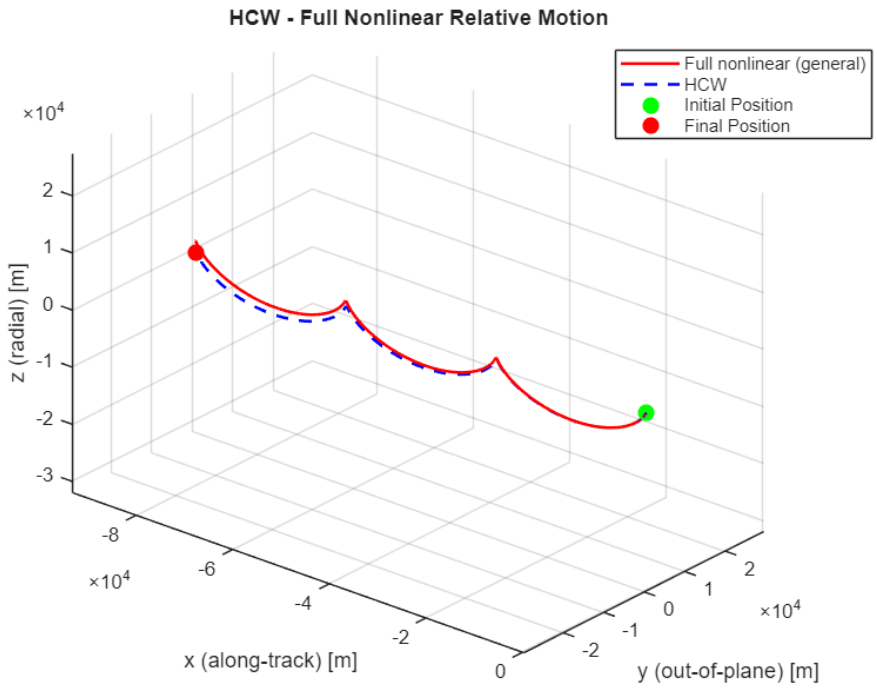


Figure 6: Comparison of the free-drift relative trajectory obtained from the full non-linear model (red) and the HCW linearisation (blue dashed).

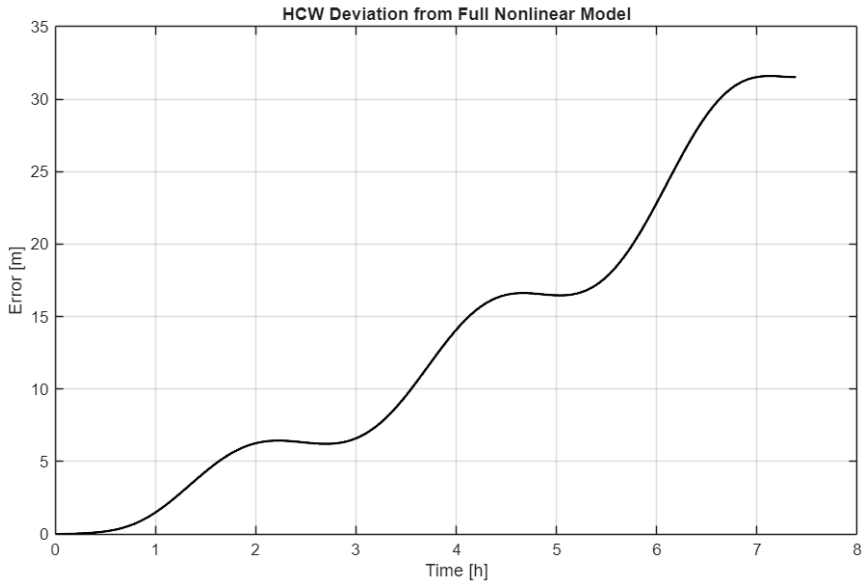


Figure 7: Position error $\varepsilon(t)$ over three orbital periods.

Quantity	HCW	Nonlinear
ΔV_{tot} [m/s]	1.601209	1.601259
$\|\Delta\mathbf{v}_1^{NL} - \Delta\mathbf{v}_1\|$ [m/s]	1.37×10^{-4}	
$\|\Delta\mathbf{v}_2^{NL} - \Delta\mathbf{v}_2\|$ [m/s]	5.3×10^{-5}	
$\|\Delta V_{tot}^{NL} - \Delta V_{tot}^{HCW}\|$ [m/s]	5.0×10^{-5}	

Table 11: Comparison of HCW and non-linear two-impulse maneuver costs, with fixed seed `rng(42)`. The differences are of order 10^{-4}–10^{-5} m/s, consistent with the linearisation error at $|\boldsymbol{\rho}|/r_{MS} < 5 \times 10^{-4}$.

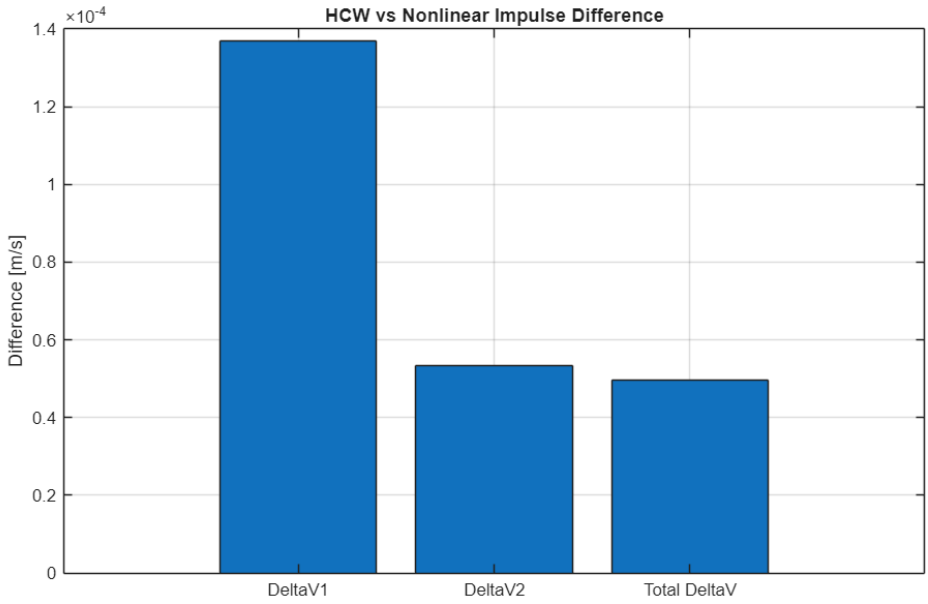


Figure 8: Absolute difference between HCW and non-linear impulse magnitudes for the two-impulse corrective maneuver. From left to right: error on $\|\Delta\mathbf{v}_1\|$, on $\|\Delta\mathbf{v}_2\|$, and on the total ΔV_{tot}. The small values confirm the validity of the HCW linearisation at the distances considered.

Quantity	HCW	Nonlinear
$\Delta V_{tot}^{(4)}$ [m/s]	0.151090	0.151090
Final pos. error [m]	1.070020	1.070017
$\|\Delta V_{tot}^{NL} - \Delta V_{tot}^{HCW}\|$ [m/s]	5.0×10^{-5}	

Table 12: V-bar approach: HCW vs. nonlinear comparison, with fixed seed `rng(42)`. The ΔV costs match to six significant figures and the final position errors differ by only 3×10^{-6} m, confirming the validity of the HCW linearisation for this maneuver phase.

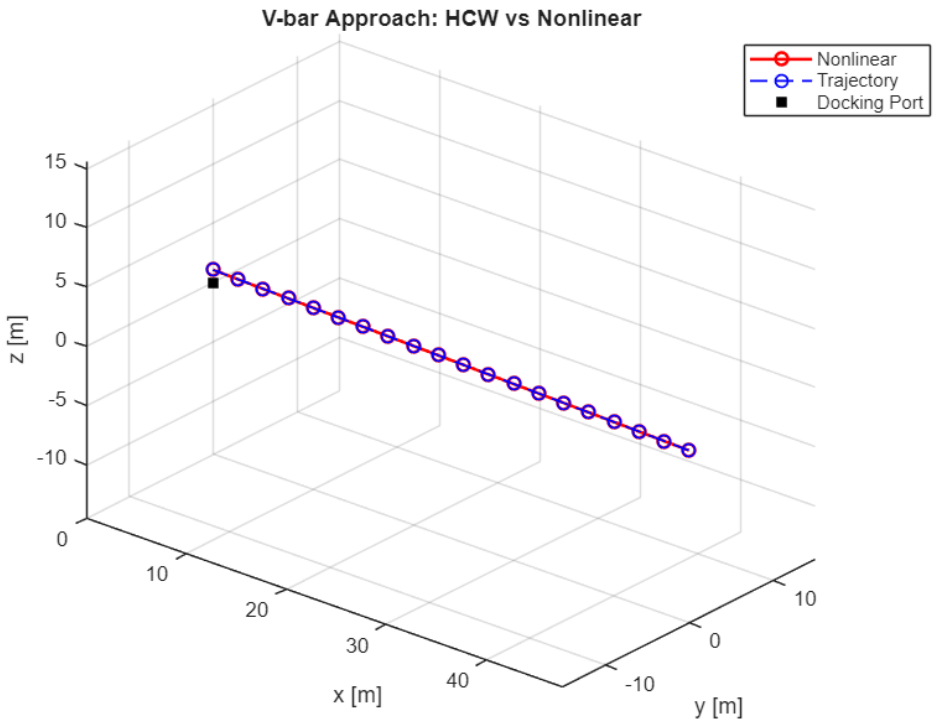


Figure 9: V-bar approach: HCW (blue) vs. nonlinear (red) trajectory comparison.

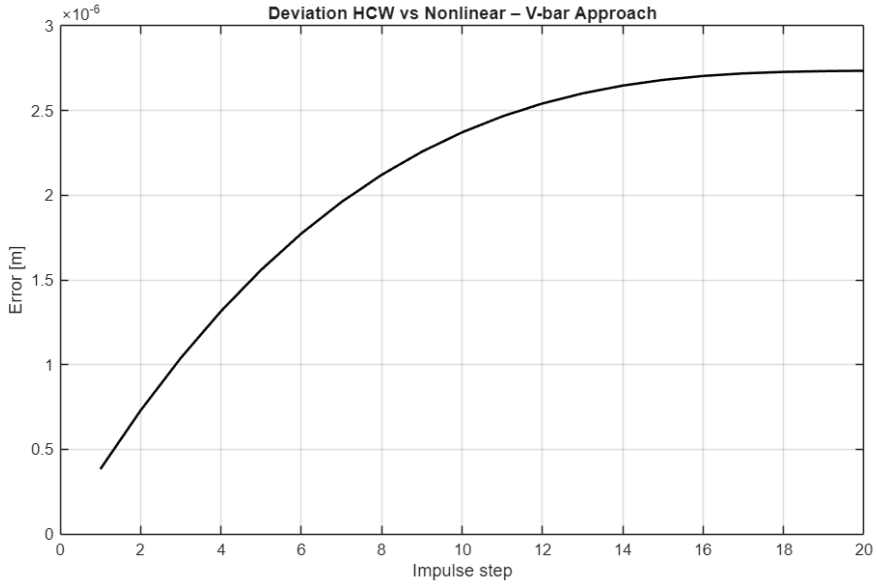


Figure 10: Step-by-step position error ε_k between the HCW and nonlinear V-bar approach trajectories over the $N_{imp} = 20$ impulse steps. The error remains at the sub-metre level throughout the approach.

3.3 Discussion

The resulting three-dimensional trajectory in the LVLH frame is shown in Figure 4. Due to the HCW coupling between the x and z channels, the relative motion exhibits periodic elliptical oscillations combined with a secular (long-term non-periodic) drift along the V-bar, whose rate is determined by the drift parameter $c = 2z_0 - \dot{x}_0/n$.

The trajectory confirms that, without corrective maneuvers, the LM would drift away from the MS on a quasi-elliptical path, making proximity operations necessary.

The computed parameters for the proximity operations are summarized in Table 9. The first impulse is stronger and sets the LM on the right course to a successful rendezvous, while the second weaker impulse cancels the LM velocity and arrests it at the 50 m hold point. The maneuver is performed in a quarter of a period.

Once at the hold point, the LM closes the distance from the docking port with a straight-line V-bar approach, composed of a series of $N_{imp} = 20$ impulses needed to keep the trajectory straight. This maneuver has total ΔV cost ten times lower than the previous one. The computed values for this maneuver are summarized in Table 10 and the trajectory is represented in Figure 5 where small radial deviations, if not adequately corrected, cause it to bring the LM to miss the docking port.

The trajectory comparison between the linear and HCW solution and the full non-linear ODE solution and error growth are shown in Figures 6 and 7. The HCW approximation is valid as long as $|\boldsymbol{\rho}| \ll r_{MS}$; given that, the initial errors are at most 1 km and $r_{MS} \approx 2137$ km, the ratio $|\boldsymbol{\rho}|/r_{MS} < 5 \times 10^{-4}$. The expected linearisation error scales as

$$\varepsilon \sim \frac{|\boldsymbol{\rho}|^2}{r_{MS}}\, n\, t, \tag{60}$$

which over three orbital periods ($3T \approx 7.4$ h) yields an estimate of order 10^1 m, consistent with the numerical result and confirming that the linear model is accurate over the time span considered (Figure 7). It is important to note that the error norm between the differently obtained values lies around the $10^{-4} \div 10^{-5}$ order of magnitude: given the small relative distances (< 1 km), this difference is expected to be well below 0.1 m/s, as reported in Table 11. The V-bar straight-line approach also presents the same order of the error norm, confirming that the HCW linearisation holds true and valid for the considered framework (Table 12). Again, the non-linear trajectory (Figure 9) is affected by radial deviation: at each step, the deviation becomes smaller and the error remains under the metre throughout the approach (Figure 10).

4 Part 3 - Perturbed Orbital Dynamics

Analytic and numeric re-computation of the entire maneuver within the Perturbed Orbital Dynamics framework, a more realistic approach to the real and different environment of the Moon. The following sections analyse the effect of relevant orbital perturbations on the LM's orbit and assess their impact on the rendezvous mission.

4.1 Methodology

Most relevant perturbations in LLO Considering a spacecraft in Low Lunar Orbit (LLO) at the altitude of h equal to a range from 100 to 400 kilometres, the dominant perturbations differ significantly from those in the Earth counterpart (Low Earth Orbit, LEO) because the Moon has:

- A weak gravity field
- Strong gravitational irregularities caused by mascons (mass concentrations)
- No atmosphere, hence no drag
- Relatively large third-body effect from Earth

We can simply identify the nominal central gravitational acceleration using the formula:

$$a_0 = \frac{\mu_M}{r^2}$$

- At $h = 100\,\text{km}$: $a_0 \approx 1.45\,\text{m/s}^2$
- At $h = 400\,\text{km}$: $a_0 \approx 1.07\,\text{m/s}^2$

We can now study the most relevant perturbations acting on a spacecraft in LLO.

Lunar Gravity Field Non-Uniformity The dominant perturbation is due to the lunar highly irregular gravity-field due to mass concentrations, called mascons. Consequently, low-degree harmonics such as J_2 and tesseral terms like C_{22} are much more important in LLO than in LEO. The acceleration associated with the oblateness coefficient J_2 scales as

$$a_{J_2} \sim \frac{3}{2} J_2 \frac{\mu_M R_M^2}{r^4}.$$

Using $J_2 \approx 2.03 \times 10^{-4}$, one obtains an order-of-magnitude estimate $a_{J_2} \sim 10^{-4}$ m/s^2. The physical effects that J_2 produces are: nodal precession, apsidal precession and a long-term orbital plane revolution.
However, unlike Earth, the Moon's J_2 is not the only important harmonic. As a matter of fact, the lunar term C_{22} is unusually large:

$$C_{22} \approx 2.2 \times 10^{-5},$$

The associated acceleration magnitude is approximately

$$a_{C_{22}} \sim 10^{-5}\text{–}10^{-4}\ \text{m/s}^2.$$

This is usually the dominant perturbation for low-altitude lunar missions below 200 km, as mascon-induced tesseral harmonics can strongly destabilize the orbit and eventually lead to impact trajectories if no station-keeping maneuvers are performed.

Third-Body Perturbations Another significant perturbation affecting satellites around the Moon is the third-body perturbation from Earth. The tidal acceleration can be estimated as

$$a_{3B,E} \sim \mu_E \frac{r}{d^3},$$

where $\mu_E = 3.986 \times 10^{14}$ m^3/s^2 and $d \approx 3.84 \times 10^8$ m is the Earth-Moon distance.

This yields

$$a_{3B,E} \sim 10^{-5} \text{ m/s}^2.$$

Earth third-body perturbations produce:

- long-term eccentricity variations,
- inclination changes,
- orbital plane perturbations.

Although weaker than the strongest mascon perturbations, Earth gravity remains one of the primary contributors to long-term orbital evolution in LLO.
Another type of third-body perturbation is due to the presence of the Sun. The solar third-body acceleration is similarly estimated:

$$a_{3B,S} \sim \mu_S \frac{r}{d_S^3},$$

with $\mu_S = 1.327 \times 10^{20}$ m^3/s^2, $d_S \approx 1.5 \times 10^{11}$ m.

The resulting order of magnitude is

$$a_{3B,S} \sim 10^{-8}\text{–}10^{-7} \text{ m/s}^2.$$

Its influence is similar to the terrestrial one but generally secondary compared with Earth third-body effects and lunar gravity harmonics.

Solar Radiation Pressure The last major perturbation in LLO is the solar radiation pressure (SRP) which produces an acceleration approximately given by

$$a_{SRP} = P_{SR} C_R \frac{A}{m},$$

where $P_{SR} \approx 4.5 \times 10^{-6}$ N/m^2 is the solar radiation pressure at 1 AU, $C_R \approx 1.2 \div 1.5$ is the reflectivity coefficient, A/m is the spacecraft area-to-mass ratio.

For a typical lunar orbiter,

$$\frac{A}{m} \sim 0.01 \text{ m}^2/\text{kg},$$

leading to

$$a_{SRP} \sim 10^{-8}\text{–}10^{-7} \text{ m/s}^2.$$

Its effects include a slow secular drift and an eccentricity growth. They are smaller than gravity harmonics and Earth perturbations but may become relevant for very lightweight spacecraft and missions requiring precise navigation.

A summary of the most relevant perturbations in LLO is reported in Table 13.

Perturbation	Typical Acceleration
Lunar gravity harmonics (J_2, C_{22}, mascons)	10^{-5}–10^{-4} m/s^2
Earth third-body	10^{-6}–10^{-5} m/s^2
Solar third-body	10^{-8}–10^{-7} m/s^2
Solar radiation pressure	10^{-8}–10^{-7} m/s^2

Table 13: Most relevant perturbation sources in LLO and their respective acceleration contribution

Numerical propagation of the J_2 perturbed orbit The J_2 perturbation is implemented using the *Cowell method*, which integrates the full equations of motion including the perturbing acceleration directly in Cartesian coordinates:

$$\ddot{\mathbf{r}} = -\frac{\mu_M}{r^3}\mathbf{r} + \mathbf{a}_{J_2}, \tag{61}$$

where the J_2 perturbing acceleration in the inertial frame is given by [2]:

$$a_{J_2,x} = -\frac{3}{2}\frac{\mu_M J_2 R_M^2}{r^5} x\left(1 - 5\frac{z^2}{r^2}\right), \tag{62}$$

$$a_{J_2,y} = -\frac{3}{2}\frac{\mu_M J_2 R_M^2}{r^5} y\left(1 - 5\frac{z^2}{r^2}\right), \tag{63}$$

$$a_{J_2,z} = -\frac{3}{2}\frac{\mu_M J_2 R_M^2}{r^5} z\left(3 - 5\frac{z^2}{r^2}\right), \tag{64}$$

with $J_2^{\text{Moon}} = 2.033 \times 10^{-4}$ and $R_M = 1737.4$ km.

The Cowell method is preferred over the Gauss variational equations because the mission orbits are circular ($e = 0$) and equatorial ($i = 0$), which produces singularities in the Gauss formulation. Although Cowell requires smaller integration steps and accumulates round-off errors more rapidly on long propagations, it is the more robust choice for this scenario.

Both spacecraft are propagated from $t = 0$ to $t = t_{\text{wait}} + \Delta t$ (total duration ≈ 10.75 h) with two separate integrations:

- **Keplerian**: only the two-body gravitational acceleration.
- **J2-perturbed**: two-body acceleration plus $\mathbf{a}_{J_2}$.

The impulsive burn ΔV_1 is applied at the end of the waiting phase in the prograde direction, consistently with Part 2. Both integrations use `ode45` with `RelTol` $= 10^{-12}$ and `AbsTol` $= 10^{-14}$.

At each time step, the classical orbital elements are computed from the Cartesian state $(\mathbf{r}, \mathbf{v})$ using the following relations:

$$a = -\frac{\mu_M}{2\varepsilon}, \qquad \varepsilon = \frac{v^2}{2} - \frac{\mu_M}{r}, \tag{65}$$

$$\mathbf{h} = \mathbf{r} \times \mathbf{v}, \tag{66}$$

$$\mathbf{e} = \frac{1}{\mu_M}\left[\left(v^2 - \frac{\mu_M}{r}\right)\mathbf{r} - (\mathbf{r}\cdot\mathbf{v})\,\mathbf{v}\right], \tag{67}$$

$$\mathbf{N} = \hat{k} \times \mathbf{h}, \tag{68}$$

$$i = \arccos\left(\frac{h_z}{\|\mathbf{h}\|}\right), \tag{69}$$

$$\Omega = \arccos\left(\frac{N_x}{\|\mathbf{N}\|}\right), \tag{70}$$

$$\omega = \arccos\left(\frac{\mathbf{N}\cdot\mathbf{e}}{\|\mathbf{N}\|\,\|\mathbf{e}\|}\right), \tag{71}$$

$$\nu = \arccos\left(\frac{\mathbf{e}\cdot\mathbf{r}}{\|\mathbf{e}\|\,\|\mathbf{r}\|}\right), \tag{72}$$

where $\mathbf{N} = \hat{k}\times\mathbf{h}$ is the node vector, $\hat{k} = [0,0,1]^\top$ is the polar axis unit vector, and the quadrant ambiguity in Ω, ω, and ν is resolved by checking the sign of the relevant component (N_y, e_z, and $\mathbf{r}\cdot\mathbf{v}$ respectively).

Numerical propagation of the J_2 + third-body (Earth) perturbed orbit The Earth's gravitational perturbation on a lunar orbiter is modelled as a third-body acceleration:

$$\mathbf{a}_{3B} = \mu_\oplus\left(\frac{\mathbf{r}_\oplus - \mathbf{r}}{\|\mathbf{r}_\oplus - \mathbf{r}\|^3} - \frac{\mathbf{r}_\oplus}{\|\mathbf{r}_\oplus\|^3}\right), \tag{73}$$

where $\mu_\oplus$ is the Earth's gravitational parameter and is equal to 3.986×10^5 km^3/s^2, $\mathbf{r}$ is the satellite position with respect to the Moon, and $\mathbf{r}_\oplus$ is the Earth position with respect to the Moon, modelled as a circular orbit at the mean Earth-Moon distance d_{EM} equal to 384 400 km:

$$\mathbf{r}_\oplus(t) = d_{EM}\begin{bmatrix}\cos(n_E t)\\ \sin(n_E t)\\ 0\end{bmatrix}, \qquad n_E = \sqrt{\frac{\mu_\oplus + \mu_M}{d_{EM}^3}}. \tag{74}$$

The combined J_2 + third-body ODE is integrated with Cowell's method using the same `ode45` tolerances as the previous propagation (`RelTol` $= 10^{-12}$ and `AbsTol` $= 10^{-14}$). The reference system considered for this phase is a selenocentric reference system, centred in the Moon's Centre of Mass.

The relative position drift between LM and MS induced by the combined perturbations is defined as

$$\Delta\rho = \left\| \left(\mathbf{r}_{\mathrm{LM}}^{J_2+3B} - \mathbf{r}_{\mathrm{MS}}^{J_2+3B}\right) - \left(\mathbf{r}_{\mathrm{LM}}^{\mathrm{Kep}} - \mathbf{r}_{\mathrm{MS}}^{\mathrm{Kep}}\right)\right\|. \tag{75}$$

4.2 Results

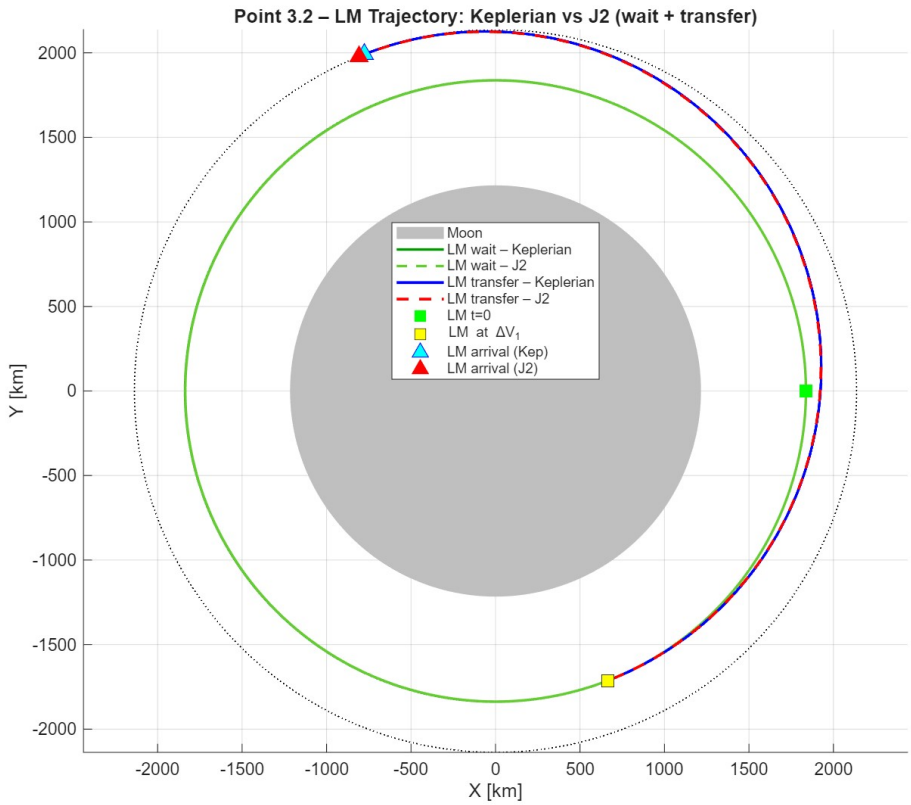


Figure 11: LM trajectory: Keplerian (blue) vs. J_2-perturbed (red dashed) over the full maneuver (waiting phase + Hohmann transfer). The yellow square marks the ΔV_1 burn epoch; the coloured triangles mark the respective arrival points.

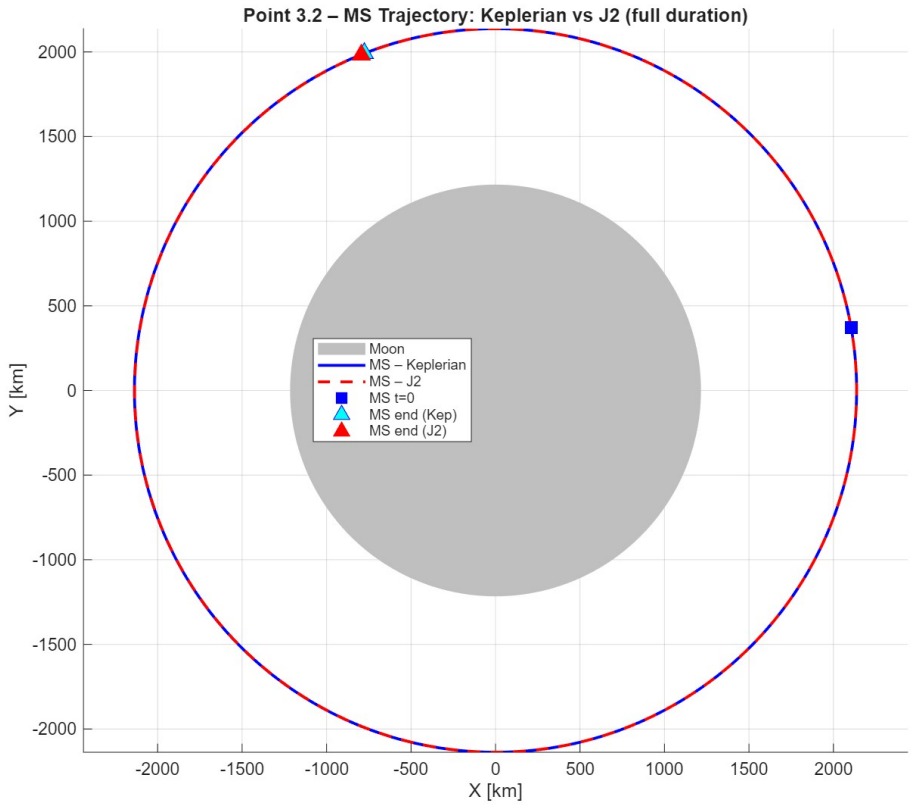


Figure 12: MS trajectory: Keplerian (blue) vs. J_2-perturbed (red dashed) over the full mission duration.

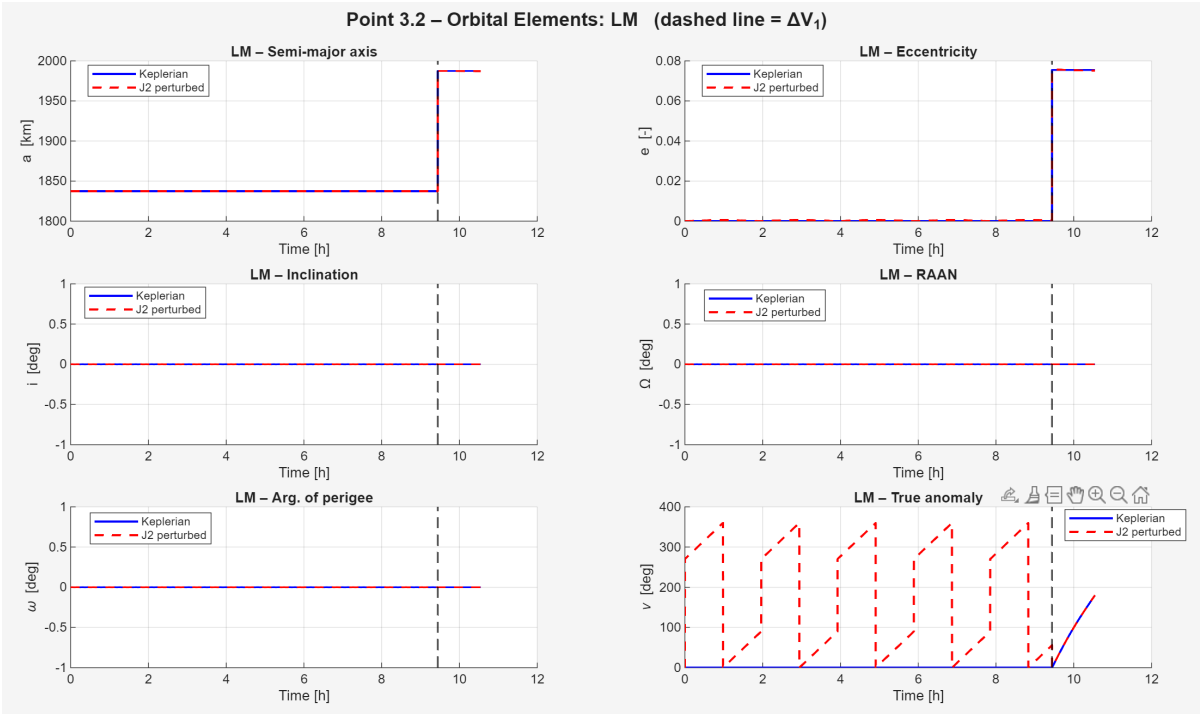


Figure 13: Orbital elements of the LM: Keplerian (blue) vs. J_2-perturbed (red dashed). The vertical dashed line marks the ΔV_1 burn epoch. The true anomaly ν shows the most significant drift, while i, Ω, and ω remain zero due to the equatorial circular orbit.

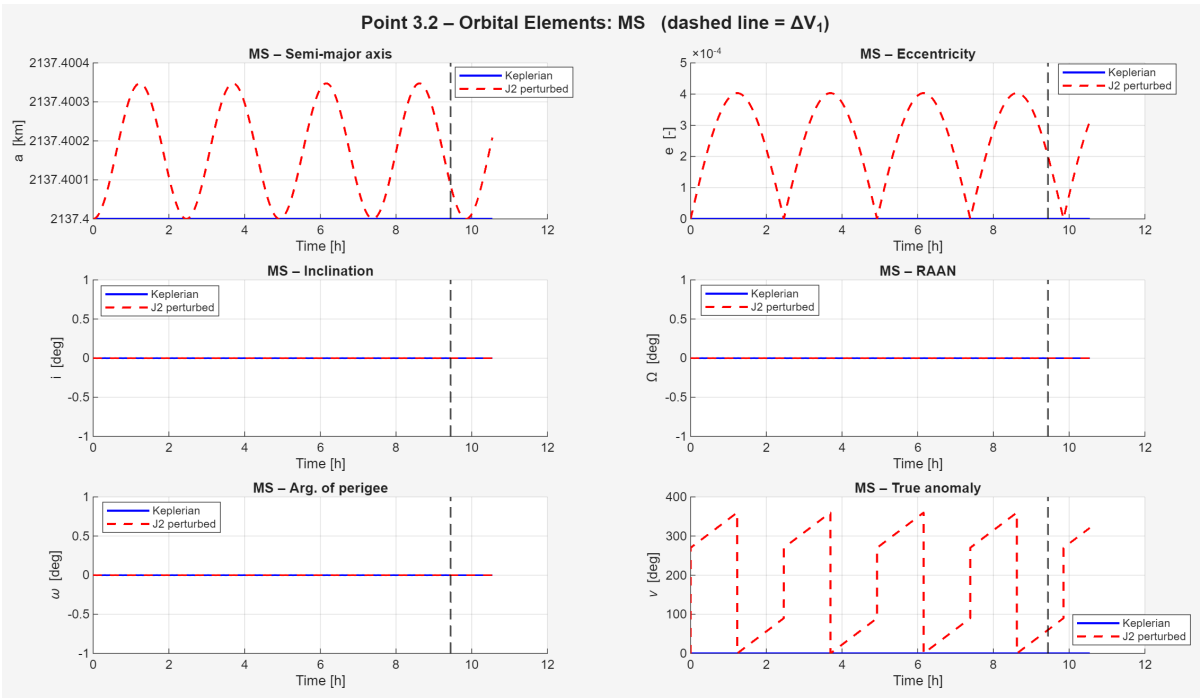


Figure 14: Orbital elements of the MS: Keplerian (blue) vs. J_2-perturbed (red dashed). The vertical dashed line marks the ΔV_1 burn epoch.

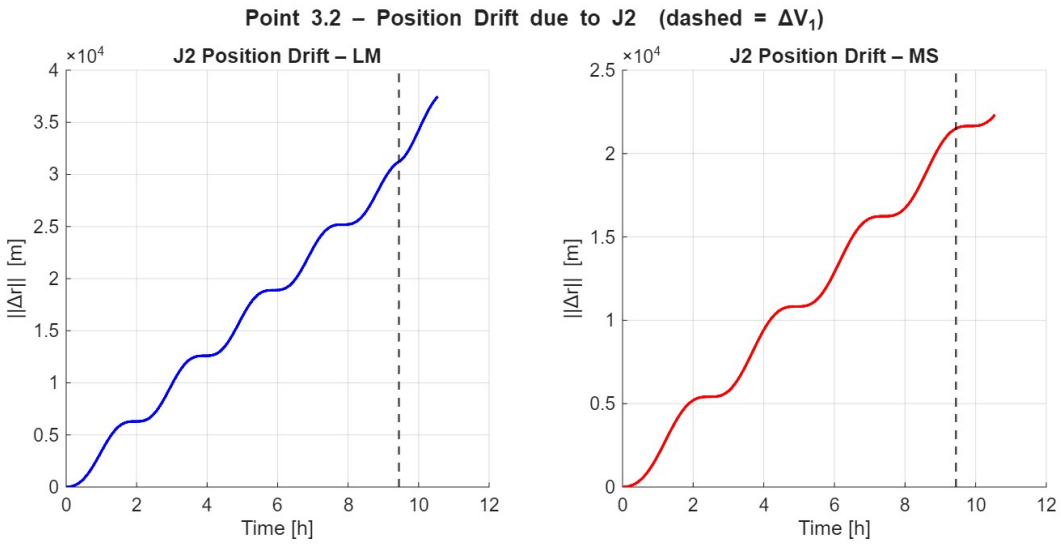


Figure 15: Absolute position drift due to perturbation for the LM (left) and MS (right). The vertical dashed line marks the ΔV_1 burn epoch.

S/C	Epoch	Δa [km]	Δe [-]	Δi [deg]	$\Delta\Omega$ [deg]	$\Delta\omega$ [deg]	$\Delta\nu$ [deg]
LM	End of wait	1.72×10^{-4}	3.06×10^{-4}	0	0	0	55.88
LM	Rendezvous	-0.115	-3.01×10^{-4}	0	0	0	0.196
MS	End of wait	8.6×10^{-5}	2.01×10^{-4}	0	0	0	60.08
MS	Rendezvous	2.08×10^{-4}	3.12×10^{-4}	0	0	0	320.78

Table 14: Orbital element drift ($\Delta\text{OE} = \text{OE}_{J_2} - \text{OE}_{\text{Kep}}$) at the end of the waiting phase and at the final rendezvous epoch.

Spacecraft	Drift at end of wait [m]	Drift at rendezvous [m]	Max drift [m]
LM	31 219.91	37 477.15	37 477.15
MS	21 496.09	22 337.24	22 337.24

Table 15: Absolute position drift due to J2 perturbation ($\|\mathbf{r}_{J_2} - \mathbf{r}_{\text{Kep}}\|$) at key mission epochs.

Quantity	Value	Unit
Relative drift $\Delta\rho$ at rendezvous (J_2+3B vs Kep)	15.02	km
Max LM absolute drift (J2+3B vs Kep)	38 177.0	m
Max MS absolute drift (J2+3B vs Kep)	23 152.8	m
Max LM–MS relative drift over full maneuver	15 024.2	m
Initial rendezvous errors (Section 3)	500–1000	m
Amplification factor $\Delta\rho \,/\, \delta r_0$	$\sim 1.5 \times 10^4$	–

Table 16: Position drift results for the combined J_2 and Earth third-body perturbation over the full maneuver duration (≈ 10.75 h).

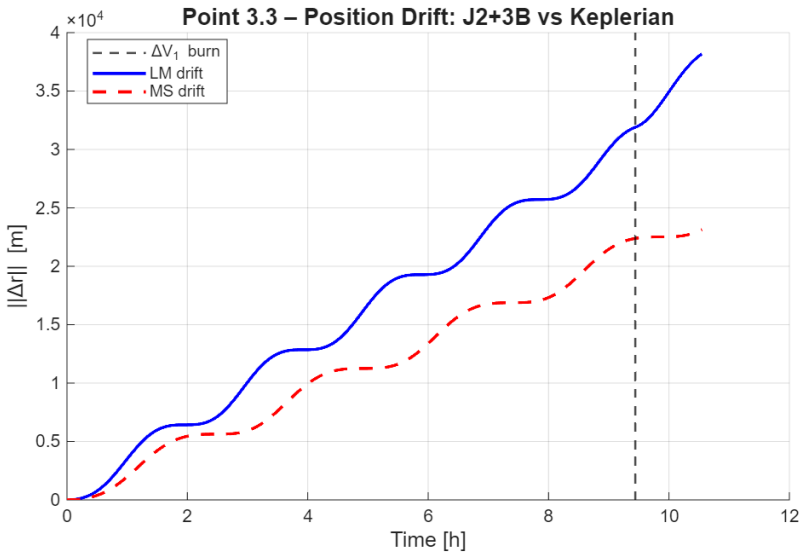


Figure 16: Absolute position drift of LM (blue) and MS (red dashed) with respect to the Keplerian reference, under the combined J_2 and Earth third-body perturbation. The vertical dashed line marks the ΔV_1 burn epoch.

4.3 Discussion

Figures 11 and 12 show the Keplerian and J2-perturbed trajectories for the LM and MS respectively. The perturbed trajectories visibly deviate from the ideal Keplerian arcs, with the LM showing a larger displacement due to its lower orbit altitude ($h_1 = 100$ km) and correspondingly stronger J2 acceleration.

During the motion, the trajectory perturbations also imply the orbital elements evolution in time: Figures 13 and 14 show the time evolution of the six classical orbital elements for the LM and MS under both the Keplerian and J2-perturbed models.

The orbits are equatorial (i is equal to 0) and circular (e is equal to 0), so the RAAN (Ω) and the argument of perigee ω lose their geometrical meaning. If we try to compute them nonetheless, they give a numerical singularity. A solution to this problem, as suggested by various texts ([2], [10]) could be to use equinoctial elements (mathematically complex) or to force RAAN and ω to zero when i and e are getting smaller and are in the neighbourhood of zero.
We decided to use the latter solution, and force them to zero in order to close the system. In fact, in Figures 13 and 14 it's possible to see that the RAAN and ω are nominally zero due to the previous assumptions and remain unaffected by J2 in the equatorial plane (the J2 acceleration has no out-of-plane component for $i = 0$). The semi-major axis a and eccentricity e show small oscillations induced by the perturbation, while the true anomaly ν diverges progressively between the two models, reflecting the cumulative phase drift.

The absolute position drift $\|\Delta\mathbf{r}\| = \|\mathbf{r}_{J_2} - \mathbf{r}_{\text{Kep}}\|$ is represented in Figure 15 for each spacecraft as a function of time. The numerical results are summarised in Table 14 for the orbital element drift and in Table 15 for the absolute position drift. The LM drift (∼37 km at rendezvous) is larger than the MS drift (∼22 km) because the LM orbits at lower altitude where the J2 acceleration is stronger. The large true anomaly drift at the end of the waiting phase ($\Delta\nu_{\text{LM}} \approx 55.9°$, $\Delta\nu_{\text{MS}} \approx 60.1°$) reflects the cumulative phase error accumulated over the ∼9.44 h waiting time. These drifts are one to two orders of magnitude

larger than the initial rendezvous errors defined in Section 3 (500-1000 m), confirming that J2 has a significant impact on the mission and needs to be considered to have acceptable accuracy.

When considering also the third-body perturbations, we can compare the position error with the random error imposed in Section 3. The initial relative errors introduced in Section 3 (Part 2) were in the range 500-1000 m. The relative drift induced by J_2 and third-body perturbations over the full duration of the maneuver (≈ 10.75 h) is ~ 15 km (Table 16), which is nearly two orders of magnitude larger than those errors. This drift is therefore operationally significant: it would completely overwhelm the initial rendezvous errors of Section 3 and must be accounted for in any realistic mission design.

The results in Table 16 show that the J_2 and third-body perturbations induce position drifts of tens of kilometres over the duration of the mission, far exceeding the initial rendezvous errors considered in Section 3. This has direct implications for the transfer strategy and the rendezvous timeline.
The largest drift accumulates during the waiting phase (~ 9.44 h), during which both spacecraft orbit freely without correction. The true anomaly drift of $\sim 56°$ for the LM and $\sim 60°$ for the MS means that by the time the Hohmann transfer begins, the spacecraft are no longer in the relative geometry computed by the ideal Keplerian analysis. A rendezvous designed purely on the Keplerian phasing would fail by tens of kilometres.

Two main approaches can be adopted to mitigate the perturbation effects:

- **Phasing recalculation.** Before executing ΔV_1, the actual states of both spacecraft are measured and a corrected phasing angle and waiting time are computed, accounting for the perturbation-induced drift. This requires onboard navigation and recomputation of the transfer window, but does not require additional propellant beyond a small correction burn.
- **Intermediate corrective maneuvers.** A sequence of small station-keeping burns can be applied during the waiting phase to keep the LM on the nominal Keplerian trajectory. The required ΔV for station-keeping against J2 scales as $\Delta V_{SK} \sim a_{J_2}\,\Delta t$; for $a_{J_2} \sim 10^{-4}$ m/s^2 and $\Delta t \approx 35{,}000$ s this gives $\Delta V_{SK} \sim 3.5$ m/s, which is comparable to the Hohmann transfer cost (0.12 km/s) but non-negligible over multiple corrections.

The accumulated relative drift of ~ 15 km at the end of the transfer must be corrected before proximity operations can begin. Using the HCW framework of Section 3, the two-impulse corrective maneuver cost scales as

$$\Delta V_{\text{corr}} \sim \frac{n\,\Delta\rho}{2}, \tag{76}$$

where n is the mean motion of the MS orbit.
With $n \approx 7.09 \times 10^{-4}$ rad/s and $\Delta\rho = 15$ km:

$$\Delta V_{\text{corr}} \sim \frac{7.09 \times 10^{-4} \times 15}{2} \approx 5.9 \text{ m/s}. \tag{77}$$

This is about 3.7 times larger than the corrective ΔV computed in Section 3 for the 500-1000 m initial errors, and would represent a significant addition to the mission propellant budget. These considerations suggest that either real-time navigation and phasing update, or a short waiting phase (minimising drift accumulation), should be preferred over passive waiting followed by a large correction.

5 Part 4 - Robotic Berthing - Manipulator Modelling and Dynamics

Dynamics of a robotic manipulator designed for berthing operation between the MS and the LM. The arm is modelled as an open-chain multibody system with 5 revolute joints. Forward kinematics is used to compute the pose of each link and of the end-effector. Differential kinematics is then formulated through the System Jacobian matrix N, allowing the propagation of velocities, wrenches and generalized forces throughout the structure.

5.1 Methodology

The robotic manipulator arm is installed on the Mothership and performs the berthing operations with the Lunar Module. The reference spacecraft used for the Mothership is the Tianhe Module of the Tiangong Space Station, the Chinese Space Station. The module orbits around earth at an altitude of 400 km, and our MS orbits around the moon at the same altitude. It has a gross mass of about 22,000 kg, a length of 19.1 m and a diameter of 4.2 m. and is equipped with a robotic arm with a 10 m operational space. For what concerns our mission design, we will consider a robotic manipulator with an operational space of 8 m. In Table 17 are listed the length and mass properties of the links.

Link	**Length [m]**	**Mass [kg]**	I_{xx} **[kg m^2]**	I_{yy} **[kg m^2]**	I_{zz} **[kg m^2]**
0	19.1 x 4.2	22000	-	-	-
1	2.5	55.55	29.1166	29.1166	0.3673
2	2	54.44	18.3270	18.3270	0.3600
3	1.5	53.33	10.1758	10.1758	0.3526
4	1.2	52.66	6.4939	6.4939	0.3482
5	0.8	51.78	2.9326	2.9326	0.3424

Table 17: Mass properties of the robotic manipulator's links (Assumptions)

These calculations are obtained considering the manipulator's structure as thin-walled cylinders in carbon fiber composite with epoxy matrix, where the density ρ is equal to 1550 kg/m^3. The wall thickness used is equal to 0.002 m. For each link, we have added an estimated mass of 50 kg in addition to the structural mass, accounting for the mass of the electrical systems (cables and motors) and systems for other ancillary functions (cooling, radiation protection, etc.). The end-effector, with the grasping mechanism necessary for the berthing operations, is estimated to be about 250 kg; that adds up to a total robotic manipulator system mass of around 517.76 kg. The joint mass and dimensions are neglected in the following computation in order to simplify the modelling of the reference systems for the various links. Therefore, the joints are considered ideal.
The Lunar Module is modelled following the Blue Ghost MK1 spacecraft by Firefly Aerospace: it is 2 m tall and 3.5 m wide, with a dry mass of approximately 469 kg. This evaluation do not consider the tanks and engine mass needed to ascent from the surface: this can be estimated to be around 200 kg (dry mass), bringing the total dry mass of the LM to around 669 kg.



Figure 17: The Blue Ghost MK1 Lander [8].

Degrees of freedom Our robotic system, as can be seen from Figure 18, is composed of 5 bodies, the links, connected by five revolute joints with the base (indicated by $\mathcal{L}_0$).

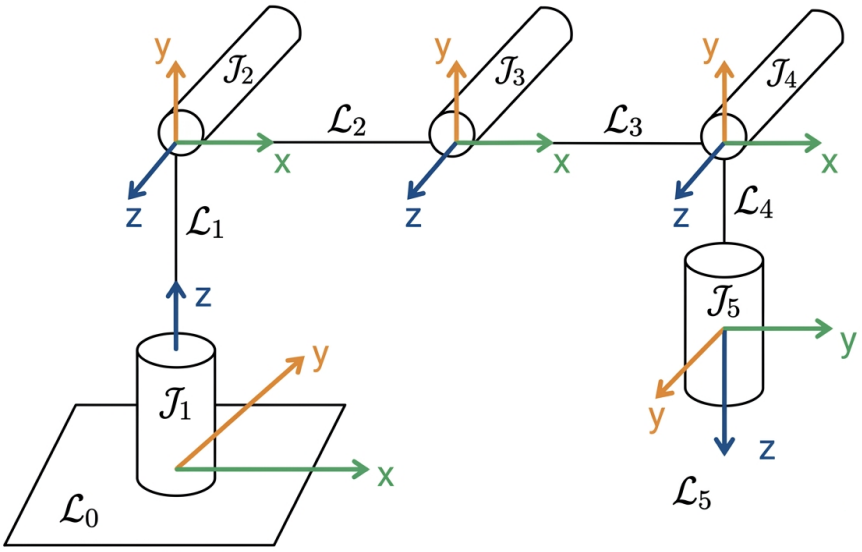


Figure 18: Schematic of the robotic arm attached to the mothership.

Following Grübler's formula,

$$m(N-1-J)+\sum f_i$$

where m is the total number of degrees of freedom in space (6 in our case), N is the number of bodies, J is the number of joints, and f_i is the number of degrees of freedom provided by the i-th joint, we can calculate the system's total number of degrees of freedom. Since all joints are revolute joints, each contributes one degree of freedom.

If we consider the robotic arm as an isolated system, the number of bodies is $N = 6$ (including the base),

while the number of joints is $J = 5$. Therefore, the term $(N-1-J)$ becomes zero, and the total number of degrees of freedom is simply equal to the sum of the joint contributions, resulting in 5 degrees of freedom.

If instead the robotic arm is considered attached to the MS, the free-floating motion of the spacecraft must also be taken into account. In this case, 6 additional degrees of freedom must be added (3 translational and 3 rotational), leading to a total of 11 degrees of freedom. In this mission we will consider this case for the system's modelling.

Forward kinematics The forward kinematics of the 5-DOF robotic manipulator is computed by composing the elementary Homogeneous Transformation Matrices (HTMs) between consecutive frames. Each matrix encodes a joint rotation followed by the rigid displacement along the subsequent link:

$$T_{EE}^{0} = T_{0}^{1}(\theta_1)\, T_{1}^{2}(\theta_2)\, T_{2}^{3}(\theta_3)\, T_{3}^{4}(\theta_4)\, T_{4}^{5}(\theta_5)\, T_{5}^{EE}, \tag{78}$$

where the superscript denotes the destination frame and the subscript the source frame. The link lengths used throughout are summarised in Table 18.

Link	**Symbol**	**Length [m]**
1	L_1	2.5
2	L_2	2.0
3	L_3	1.5
4	L_4	1.2
5	L_5	0.8
Total reach ℓ		8.0

Table 18: Link lengths of the 5-DOF robotic manipulator.

The purpose of the forward kinematics is to determine the position and orientation (the so called "pose") of the end-effector given the joint variables. Setting the out-of-plane joint angles $\theta_1 = \theta_3 = \theta_5 = 0$ and retaining θ_2 and θ_4 as the two free planar variables reduces the arm to a planar robot in the xz-plane. The resulting HTM, obtained symbolically in MATLAB and simplified, is:

$$T_{EE,2D}^{0} = \begin{bmatrix} \cos(\theta_2+\theta_4) & 0 & \sin(\theta_2+\theta_4) & (L_4+L_5)\sin(\theta_2+\theta_4) + (L_2+L_3)\cos\theta_2 \\ 0 & -1 & 0 & 0 \\ \sin(\theta_2+\theta_4) & 0 & -\cos(\theta_2+\theta_4) & L_1 + (L_2+L_3)\sin\theta_2 - (L_4+L_5)\cos(\theta_2+\theta_4) \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{79}$$

The end-effector position components read directly from the last column of (79):

$$p_x = (L_4 + L_5)\sin(\theta_2 + \theta_4) + (L_2 + L_3)\cos\theta_2, \tag{80}$$

$$p_y = 0, \tag{81}$$

$$p_z = L_1 + (L_2 + L_3)\sin\theta_2 - (L_4 + L_5)\cos(\theta_2 + \theta_4). \tag{82}$$

Two configurations are analysed to illustrate the kinematics and to verify the correctness of the HTM.

Trivial configuration. ($\theta_2 = 0$, $\theta_4 = 0$). Substituting into (80)–(82):

$$p_x = L_2 + L_3 = 3.5 \text{ m} \tag{83}$$

$$p_y = 0 \tag{84}$$

$$p_z = L_1 - L_4 - L_5 = 0.5 \text{ m} \tag{85}$$

Non-trivial configuration ($\theta_2 = \pi/4$, $\theta_4 = \pi$). Substituting $\theta_2 = \pi/4$ and $\theta_4 = \pi$ into (80)–(82):

$$p_x = (L_4 + L_5)\sin\left(\frac{5}{4}\pi\right) + (L_2 + L_3)\cos\left(\frac{\pi}{4}\right) = 1.061\,\text{m} \tag{86}$$

$$p_y = 0 \tag{87}$$

$$p_z = L_1 + (L_2 + L_3)\sin\left(\frac{\pi}{4}\right) - (L_4 + L_5)\cos\left(\frac{5}{4}\pi\right) = 6.389\,\text{m} \tag{88}$$

Differential kinematics and System Jacobian Differentiating in time, we obtain the time derivative of the position or orientation, hence the velocity (linear or angular). In this phase we use the Screw Algebra Notation introducing the twist vector, a 6×1 vector used to represent the velocity of a rigid body as an angular velocity around an axis (the joint rotation axis) and a linear velocity along this axis. Let us also define other useful quantities: Twist propagation vectors $\mathbf{p}_i$. For each revolute joint i the twist propagation vector is the 6×1 vector:

$$\mathbf{p}_i = \begin{bmatrix} \hat{z}_i \\ \hat{z}_i \times \mathbf{g}_i \end{bmatrix}, \tag{89}$$

where $\hat{z}_i = \mathbf{R}_{0i}(:, 3)$ is the joint rotation axis expressed in the inertial frame, and $\hat{z}_i \times \mathbf{g}_i$ is the linear velocity contribution of that joint to the CoM of link i.

Twist propagation matrices $\mathbf{B}_{ij}$. The 6×6 matrix that propagates the twist from the CoM of body j to the CoM of body i is:

$$\mathbf{B}_{ij} = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0} \\ \widetilde{\mathbf{r}}_{ij} & \mathbf{I}_3 \end{bmatrix}, \qquad \widetilde{\mathbf{r}}_{ij} = \text{skew}(\mathbf{C}_i - \mathbf{C}_j), \tag{90}$$

where skew$(\cdot)$ denotes the 3×3 skew-symmetric matrix operator. Five consecutive matrices are assembled: $\mathbf{B}_{10}$, $\mathbf{B}_{21}$, $\mathbf{B}_{32}$, $\mathbf{B}_{43}$, $\mathbf{B}_{54}$.

Lower-triangular propagation matrix $\mathbf{N}_l$. The matrix $\mathbf{N}_l \in \mathbb{R}^{6(n+1)\times6(n+1)}$ is assembled with 6×6 identity blocks on the diagonal and cumulative products of $\mathbf{B}$ matrices in the lower triangle:

$$\mathbf{N}_l^{(i,j)} = \begin{cases} \mathbf{I}_6 & i = j, \\ \displaystyle\prod_{k=j}^{i-1} \mathbf{B}_{k+1,k} & i > j, \\ \mathbf{0} & i < j. \end{cases} \tag{91}$$

For example, the block propagating the base twist to link 3 is $\mathbf{N}_l^{(3,0)} = \mathbf{B}_{32}\mathbf{B}_{21}\mathbf{B}_{10}$. With $n = 5$: $\mathbf{N}_l \in \mathbb{R}^{36\times36}$.

Block-diagonal input matrix $\mathbf{N}_d$. The matrix $\mathbf{N}_d \in \mathbb{R}^{6(n+1)\times(6+n)}$ separates the base twist (first 6 columns) from the joint velocity inputs (one column per joint):

$$\mathbf{N}_d = \begin{bmatrix} \mathbf{I}_6 & \mathbf{0} & & & \\ \mathbf{0} & \mathbf{p}_1 & \cdots & & \\ & & \mathbf{p}_2 & \cdots & \\ & & & \ddots & \\ & & & & \mathbf{p}_n \end{bmatrix}, \tag{92}$$

so that column $6+i$ contains $\mathbf{p}_i$ in the rows associated with body i. With $n=5$: $\mathbf{N}_d \in \mathbb{R}^{36\times 11}$.

System Jacobian $\mathbf{N}$. The full System Jacobian is obtained assembling the two block matrices:

$$\mathbf{N} = \mathbf{N}_l\, \mathbf{N}_d \in \mathbb{R}^{6(n+1)\times(6+n)}, \tag{93}$$

which, for $n=5$ (5R manipulator), yields $\rightarrow 6(5+1) = 36$ rows and $6+5=11$ columns: $\mathbf{N} \in \mathbb{R}^{36\times 11}$.

The MATLAB output confirms this:

```
Extracted dimensions of N: 36 x 11  ✓
```
(94)

Contact force propagation During the final berthing phase the robotic arm pushes the LM along the docking axis. The contact wrench is applied exclusively at the end-effector (link L_5) and is modelled as a pure force of 100 N along the inertial z-axis with no external moment:

$$\mathbf{w}_5 = \begin{bmatrix} n_x \\ n_y \\ n_z \\ f_x \\ f_y \\ f_z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 100 \end{bmatrix} \begin{bmatrix} \text{Nm} \\ \text{Nm} \\ \text{Nm} \\ \text{N} \\ \text{N} \\ \text{N} \end{bmatrix}. \tag{95}$$

This choice is physically motivated by the berthing geometry: the final approach occurs along the z-axis, and a point contact at the end-effector is assumed (no moment involved). No external loads act on any other link, so the full system wrench vector $\mathbf{w} \in \mathbb{R}^{36}$ has non-zero entries only in its last six elements.

Generalised force vector. The system-level equilibrium equation in joint space is:

$$\boldsymbol{\tau} = \mathbf{N}^T\, \mathbf{w}, \tag{96}$$

which yields the $(6+n)\times 1$ generalised force vector:

$$\boldsymbol{\tau} = \begin{bmatrix} \boldsymbol{\tau}_0 \\ \boldsymbol{\tau}_m \end{bmatrix}, \tag{97}$$

$\boldsymbol{\tau}_0 \in \mathbb{R}^6$, is the reaction wrench transmitted to the MS base and $\boldsymbol{\tau}_m \in \mathbb{R}^5$ contains the torque at each revolute joint.

The analysis is performed with 4 different arm configurations:

- Trivial $\theta_1 = \theta_2 = \theta_3 = \theta_4 = \theta_5 = 0$
- Non-trivial $\theta_1 = \theta_3 = \theta_5 = 0$, $\theta_2 = \pi/4$, $\theta_4 = \pi$
- Minimum torque $\theta_1 = \theta_3 = \theta_5 = 0$, $\theta_2 = \pi/2$, $\theta_4 = \pi/2$
- Maximum torque $\theta_1 = \theta_2 = \theta_3 = \theta_5 = 0$, $\theta_4 = \pi/2$

The third and fourth configurations are expected to generate the minimum and maximum torque, but this assumption needs to be confirmed by the calculations.

5.2 Results

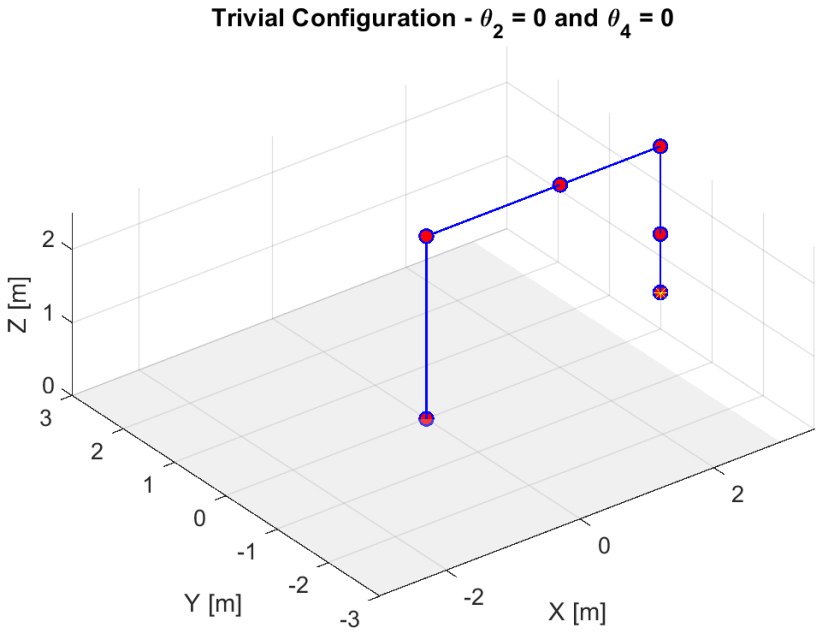


Figure 19: Trivial configuration ($\theta_i = 0$ for all i). The manipulator lies in the xz-plane. Red markers indicate the joint positions. The yellow star represents the end-effector position.

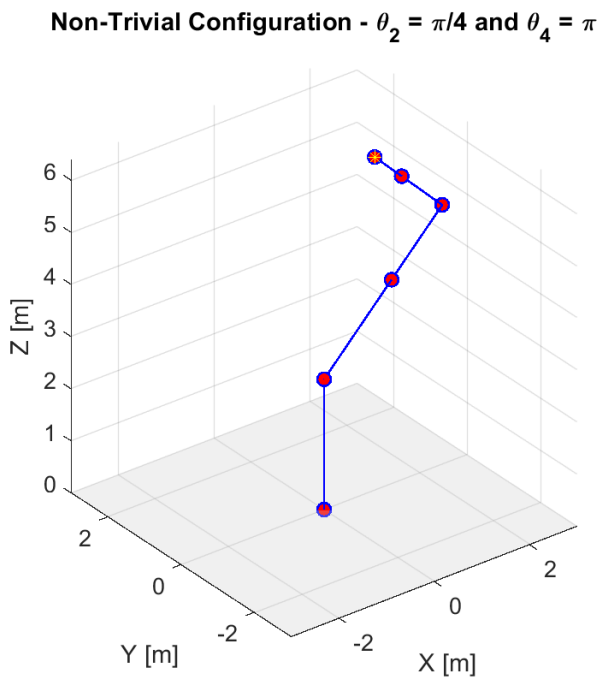


Figure 20: Non-trivial configuration ($\theta_2 = \pi/4$, $\theta_4 = \pi$). The manipulator lies in the xz-plane. Red markers indicate the joint positions. The yellow star represents the end-effector position.

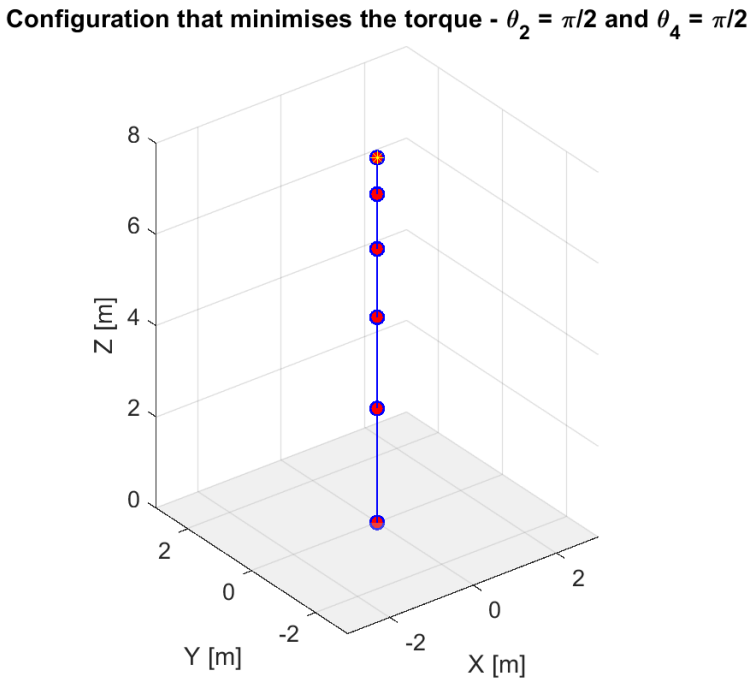


Figure 21: Minimum torque configuration ($\theta_2 = \pi/2$, $\theta_4 = \pi/2$). The manipulator lies in the xz-plane. Red markers indicate the joint positions. The yellow star represents the end-effector position.

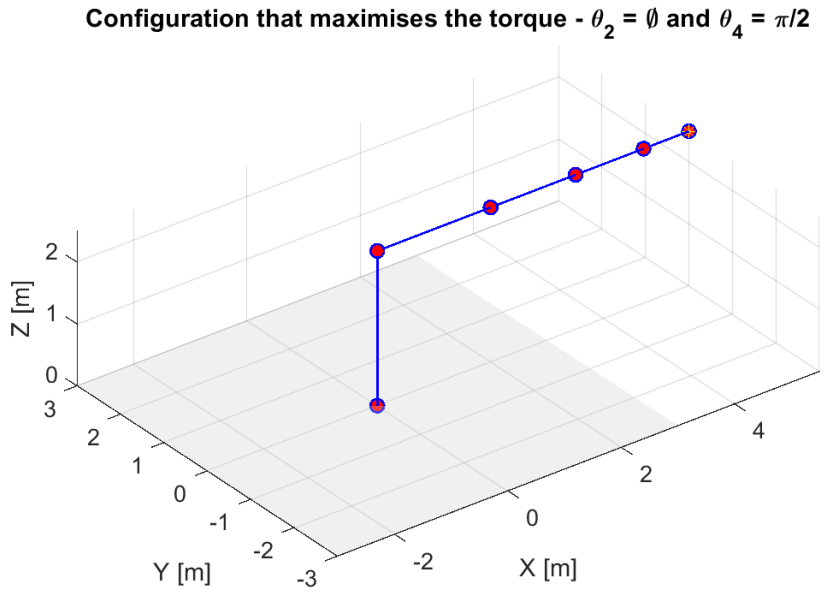


Figure 22: Maximum torque configuration ($\theta_2 = 0$, $\theta_4 = \pi/4$). The manipulator lies in the xz-plane. Red markers indicate the joint positions. The yellow star represents the end-effector position.

Configuration	n_x [Nm]	n_y [Nm]	n_z [Nm]	f_x [N]	f_y [N]	f_z [N]
Trivial (θ_2=0, θ_4=0)	0	350.00	0	0	0	100
Non-trivial (θ_2=π/4, θ_4=π)	0	134.35	0	0	0	100
Minimum torque (θ_2=π/2, θ_4=π/2)	0	0	0	0	0	100
Maximum torque (θ_2=0, θ_4=π/2)	0	510.00	0	0	0	100

Table 19: Base reaction wrench $\boldsymbol{\tau}_0 = [n_x, n_y, n_z,\ f_x, f_y, f_z]^T$ for each configuration under the contact wrench (95).

Configuration	τ_1 [Nm]	τ_2 [Nm]	τ_3 [Nm]	τ_4 [Nm]	τ_5 [Nm]
Trivial	0	−150.00	0	0	0
Non-trivial	0	7.07	113.14	28.28	0
Minimum torque	0	0	0	0	0
Maximum torque	0	−310.00	−160.00	−40.00	0

Table 20: Joint torques $\boldsymbol{\tau}_m = [\tau_1, \ldots, \tau_5]^T$ for each configuration under the contact wrench (95).

Configuration	$\sum \lvert\tau_i\rvert$ [Nm]	n_y [Nm]
Trivial	150.00	350.00
Non-trivial	148.49	134.35
Minimum torque	0	0
Maximum torque	510.00	510.00

Table 21: Cumulative joint torque demand $\sum |\tau_i|$ and base bending moment n_y for the four configurations.

5.3 Discussion

For the trivial configuration, the arm extends in the xz-plane with the end-effector at (3.5, 0, 0.5) m. The non-trivial configuration described by $\theta_2 = \pi/4$ and $\theta_4 = \pi$ leads to a three-dimensional folding of the robotic structure, which perfectly validates the analytical forward kinematics. As illustrated in the MATLAB simulation in Figure 20, the manipulator lies entirely within the xz-plane ($p_y = 0$ m), while the end-effector successfully reaches the computed spatial coordinates of $p_x = 1.061$ m (86) and $p_z = 6.389$ m (88). This rigorous graphical match fully confirms the correctness of the derived HTM.

Regarding the analysis of the contact force, from Tables 19 and 20 is possible to notice that in every configuration $f_z = 100$ N and $\tau_1 = \tau_5 = 0$: the axial force is always transmitted entirely to the base regardless of arm geometry, and the yaw joint (J_1, axis along z) and the wrist joint (J_5) produce no reaction to a pure z-force because their axes are parallel or perpendicular to the force direction with zero moment arm in their respective planes. The base moment n_y and the intermediate joint torques (τ_2-τ_4),instead, vary strongly with configuration, reflecting the changing moment arms of the contact force.

The minimum torque configuration ($\theta_2 = \pi/2$, $\theta_4 = \pi/2$) yields $\boldsymbol{\tau}_m = \mathbf{0}$ and $n_y = 0$. This result has a clear geometric interpretation: with all links aligned along the z-axis (Figure 21) the contact force $f_z = 100$ N acts *along* the arm axis, producing zero moment arm at every joint. The entire external load is transmitted directly to the base as a pure axial force with no bending moment, so no joint actuation is required to maintain equilibrium. This configuration confirms to be the minimum torque one and constitutes the ideal berthing approach direction from a structural standpoint.

The maximum torque configuration ($\theta_2 = 0$, $\theta_4 = \pi/2$) extends links from L_2 up to L_5 horizontally (Figure 22). The end-effector horizontal offset from the base is $p_x = L_2 + L_3 + L_4 + L_5 = 5.5$ m, producing the largest moment arm for the z-directed contact force. The resulting base moment $n_y = 510$ Nm and joint torques up to $|\tau_2| = 310$ Nm represent the worst-case loading on the arm. A comparison of the total joint torque demand across configurations is summarised in Table 21.

By definition, the base reaction wrench $\boldsymbol{\tau}_0$ is the net load transmitted from the robotic arm to the MS structure at the attachment point. The dominant component is the bending moment n_y (Table 19), which ranges from 0 (minimum torque configuration) to 510 Nm (maximum torque configuration). The other components are null. The axial force $f_z = 100$ N is constant and independent of configuration. For the MS attitude control system (ACS) this has two consequences. First, the bending moment n_y generates a disturbance torque on the MS body, which the ACS must counteract using thrusters or reaction wheels to prevent attitude drift during berthing. The magnitude of this disturbance, up to 510 Nm, is significant compared to typical ACS authority in low-thrust configurations, suggesting that the arm geometry should be planned to keep the moment arm small throughout the berthing sequence. Second, the constant $f_z = 100$ N pushes the MS along z, which the ACS must also compensate to maintain the

relative position with the LM.

In the case of free-floating mode for the MS, the ACS is inactive and the base reaction wrench $\boldsymbol{\tau}_0$ is no longer absorbed by the ACS but instead appears as an unbalanced external load acting on the MS. According to Newton's third law, the contact force $\boldsymbol{\tau}_0$ will cause the MS to accelerate and rotate in the opposite sense:

$$\mathbf{M}\,\dot{\boldsymbol{\nu}}_0 = -\boldsymbol{\tau}_0, \tag{98}$$

where $\mathbf{M}$ is the 6×6 mass-inertia matrix of the MS and $\boldsymbol{\nu}_0$ is its twist. Physically, the bending moment n_y will cause the MS to rotate about the y-axis, increasing the misalignment with the LM over time, while f_z will push the MS away from its nominal docking position. The arm kinematics and the MS dynamics are therefore coupled, and a free-floating analysis would require a floating-base formulation in which the MS pose is treated as an additional set of generalised coordinates. In this scenario the minimum torque configuration ($\boldsymbol{\tau}_0 = [0, 0, 0, 0, 0, 100]^T$) remains preferable because the pure axial force produces only a translational perturbation along z (no attitude disturbance), whereas the maximum torque configuration would simultaneously translate and rotate the MS, making station keeping impossible without active control.

6 Bonus Assignments

6.1 CR3BP Transfer

Repetition of the orbital transfer analysis of Section 2 in the Circular Restricted Three-Body Problem (CR3BP). Comparison of ΔV, transfer time and trajectories in the different frameworks and role of the Lagrangian points.

6.1.1 Methodology

The Circular Restricted Three-Body Problem (CR3BP) extends the two-body Keplerian model by including the gravitational influence of both the Earth and the Moon on the spacecraft, under the following assumptions [9]:

- The two massive bodies (Earth P_1 and Moon P_2) move in circular orbits about their common barycentre O.
- The distances OP_1 and OP_2 are constant, and the line segment P_1P_2 rotates about O with constant angular velocity ω.
- The mass of the spacecraft is negligible compared to the primaries.

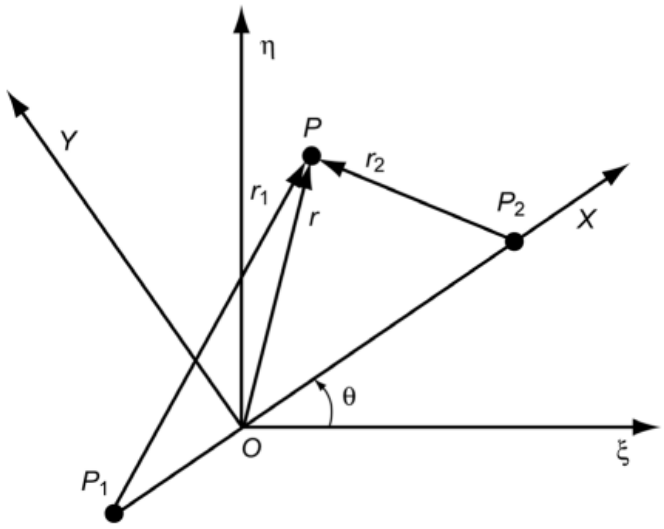


Figure 23: Inertial ($\xi\eta\zeta$) and rotating (xyz) reference frames in the CR3BP. P_1 (Earth) and P_2 (Moon) are the two primaries; O is the barycentre; P is the spacecraft. The rotating frame co-rotates with P_1P_2 at constant angular velocity ω [9].

The equation of motion of the spacecraft in the **inertial** reference frame is [9]:

$$\frac{d^2\bar{\mathbf{r}}}{dt^2} = -G\frac{m_1}{r_1^3}\bar{\mathbf{r}}_1 - G\frac{m_2}{r_2^3}\bar{\mathbf{r}}_2, \tag{99}$$

where

$$r_1^2 = (\xi-\xi_1)^2 + (\eta-\eta_1)^2 + \zeta^2, \qquad r_2^2 = (\xi-\xi_2)^2 + (\eta-\eta_2)^2 + \zeta^2 \tag{100}$$

are the squared distances from the spacecraft to Earth and Moon respectively.

If we were to use this inertial frame the mathematical formulation would be significantly more complex, because the positions of both primaries vary in time. For this reason the problem is reformulated in the **rotating** (synodic) **reference frame**, which co-rotates with the Earth–Moon system. In this frame the two primaries are fixed on the x-axis, but apparent forces appear. Substituting the transformation into (99), the equation of motion in the rotating frame becomes [9]:

$$\frac{\delta^2\bar{\mathbf{r}}}{\delta t^2} = -G\left(\frac{m_1}{r_1^3}\bar{\mathbf{r}}_1 + \frac{m_2}{r_2^3}\bar{\mathbf{r}}_2\right) - 2\bar{\boldsymbol{\omega}}\times\frac{\delta\bar{\mathbf{r}}}{\delta t} - \bar{\boldsymbol{\omega}}\times(\bar{\boldsymbol{\omega}}\times\bar{\mathbf{r}}), \tag{101}$$

where the second term is the *Coriolis acceleration* and the third is the *centrifugal acceleration*.

It is important to note that the angular momentum and energy integrals of motion of the n-body problem cannot be used for the CR3BP [9]. The reason is that the assumption that the two massive bodies move in circular orbits in a single plane is only valid if the mass of the spacecraft is exactly zero. As a consequence, conservation of energy and angular momentum is violated, and the only remaining constant of motion is the Jacobi constant.

Non-dimensional units To simplify equation (101), new units of mass, length and time are introduced. As unit of mass we take $(m_1 + m_2)$, so that the two primary masses become [9]:

$$m_1 = 1-\mu, \qquad m_2 = \mu, \tag{102}$$

where $\mu = m_2/(m_1 + m_2) = 0.01215$ is the Earth–Moon mass ratio. The distance and time units are:

$$\mathrm{DU} = d_{EM} = 384\,400 \text{ km}, \tag{103}$$

$$\mathrm{TU} = \sqrt{\frac{\mathrm{DU}^3}{\mu_{tot}}}, \qquad \mu_{tot} = \frac{\mu_M}{\mu}, \tag{104}$$

$$\mathrm{VU} = \frac{\mathrm{DU}}{\mathrm{TU}}. \tag{105}$$

In these non-dimensional units, $\omega = 1$, the Moon is located at $x_M = 1 - \mu$ from the barycentre and equation (101) reduces to:

$$\ddot{x} - 2\dot{y} = \frac{\partial U}{\partial x}, \tag{106}$$

$$\ddot{y} + 2\dot{x} = \frac{\partial U}{\partial y}, \tag{107}$$

$$\ddot{z} = \frac{\partial U}{\partial z}, \tag{108}$$

where the effective pseudo-potential is

$$U = \frac{1}{2}(x^2 + y^2) + \frac{1-\mu}{r_1} + \frac{\mu}{r_2}, \tag{109}$$

with r_1, r_2 the non-dimensional distances from the spacecraft to the Earth and Moon respectively.

At the start of the maneuver, the LM departs from the LLO ($h_1 = 100$ km) at the point where the orbit crosses the Moon-barycentre axis. In the non-dimensional rotating frame the initial position is

$$x_0 = (1-\mu) + r^*_{LM}, \quad y_0 = 0, \quad z_0 = 0, \tag{110}$$

where $(1-\mu)$ is the non-dimensional distance of the Moon from the barycentre and $r^*_{LM} = r_{LM}/\mathrm{DU}$ is the non-dimensional LLO radius.

The initial velocity is along the y-axis (prograde). To convert from the inertial frame to the rotating frame the transport theorem gives $\mathbf{v}_{rot} = \mathbf{v}_{in} - \boldsymbol{\omega} \times \mathbf{r}$. Since $\omega = 1$ in non-dimensional units:

$$\dot{y}_0 = v^*_{per} - x_0 = v^*_{per} - (1-\mu) - r^*_{LM}, \tag{111}$$

where $v^*_{per} = v_{perigee}/\mathrm{VU}$ is the non-dimensional perigee velocity of the Hohmann transfer ellipse. Numerically $(1-\mu) \approx 0.988$, so the guess used in the code is

$$\dot{y}_{0,\mathrm{guess}} = v^*_{per} - r^*_{LM}, \tag{112}$$

which omits the $(1-\mu)$ offset. Since this value is only an initial guess for the shooting method - which then adjusts $\dot{y}_0$ to satisfy the boundary conditions - the approximation does not affect the final solution.

Shooting method A shooting method is required to solve the CR3BP because it has no closed-form analytical solution for general transfer trajectories. The two-point boundary value problem (depart from LLO, arrive at MS orbit) is therefore solved numerically using the shooting method: an initial condition is "shot" forward in time using `ode45`, and `fsolve` iteratively adjusts the free variables (initial y-velocity $\dot{y}_0$ and time of flight Δt) until the terminal conditions are satisfied:

$$x(t_f) = (1-\mu) - r^*_{MS}, \qquad y(t_f) = 0. \tag{113}$$

The solver converges to exit flag = 1 with tolerances `TolFun` $= 10^{-10}$ and `TolX` $= 10^{-10}$.

Jacobi constant In this analysis, we cannot compare the obtained results against an analytical reference because, as previously mentioned, the three-body problem has no general closed-form solution [9]. The only verification available is the conservation of the Jacobi constant.

Unlike the two-body problem, which has three independent first integrals (ε, $\mathbf{h}$, $\mathbf{e}$), the CR3BP possesses only one constant of motion: the Jacobi constant [9]

$$C = 2U - V^2 = x^2 + y^2 + \frac{2(1-\mu)}{r_1} + \frac{2\mu}{r_2} - \left(\dot{x}^2 + \dot{y}^2 + \dot{z}^2\right), \tag{114}$$

where $V^2 = \dot{x}^2 + \dot{y}^2 + \dot{z}^2$ is the squared velocity in the rotating frame and U is the effective pseudo-potential defined above. C is exactly conserved along every solution of the CR3BP equations of motion and plays a role analogous to the specific orbital energy in the two-body problem.

6.1.2 Results

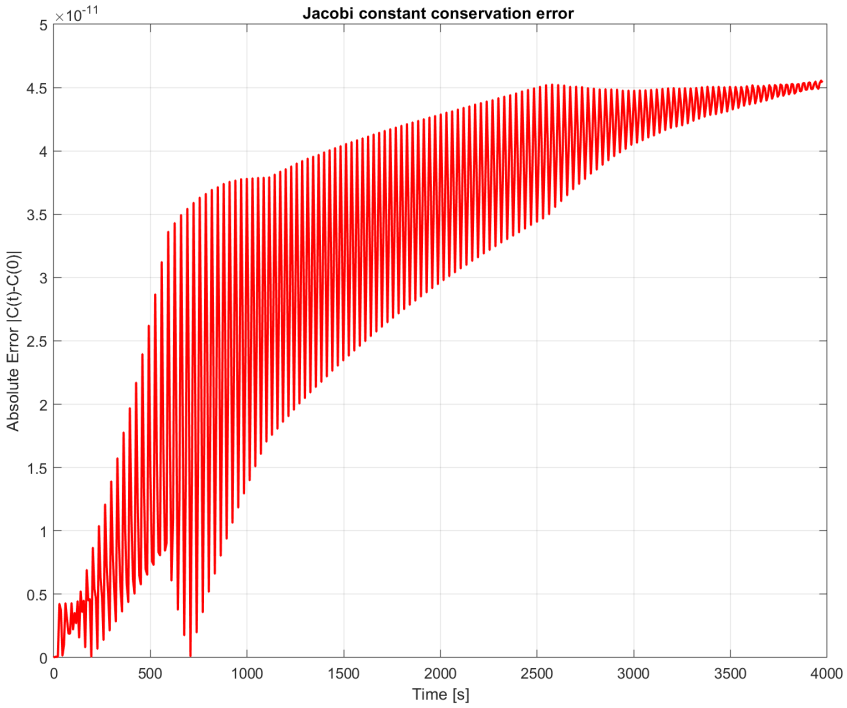


Figure 24: Absolute error of the Jacobi constant $|C(t) - C(0)|$ along the CR3BP trajectory. The error remains below 5×10^{-11}, confirming the accuracy of the integration. The oscillatory pattern is numerical noise from the adaptive integrator, not a physical effect.

Quantity	Two-body Hohmann	CR3BP
ΔV_1 [km/s]	0.0605	0.0605
ΔV_2 [km/s]	0.0583	0.0583
ΔV_{tot} [km/s]	0.1188	0.1188
Time of flight [s]	3975.17	3990.84
Δ(ToF) [s]	+15.67	
Target orbit radius [km]	2137.400	
Achieved radius (CR3BP) [km]	2137.325	
Radial error [m]	74.77	
Max Jacobi error $\lvert C(t) - C(0)\rvert$	4.56×10^{-11}	

Table 22: Comparison of the two-body Hohmann transfer and the CR3BP trajectory obtained via the shooting method.

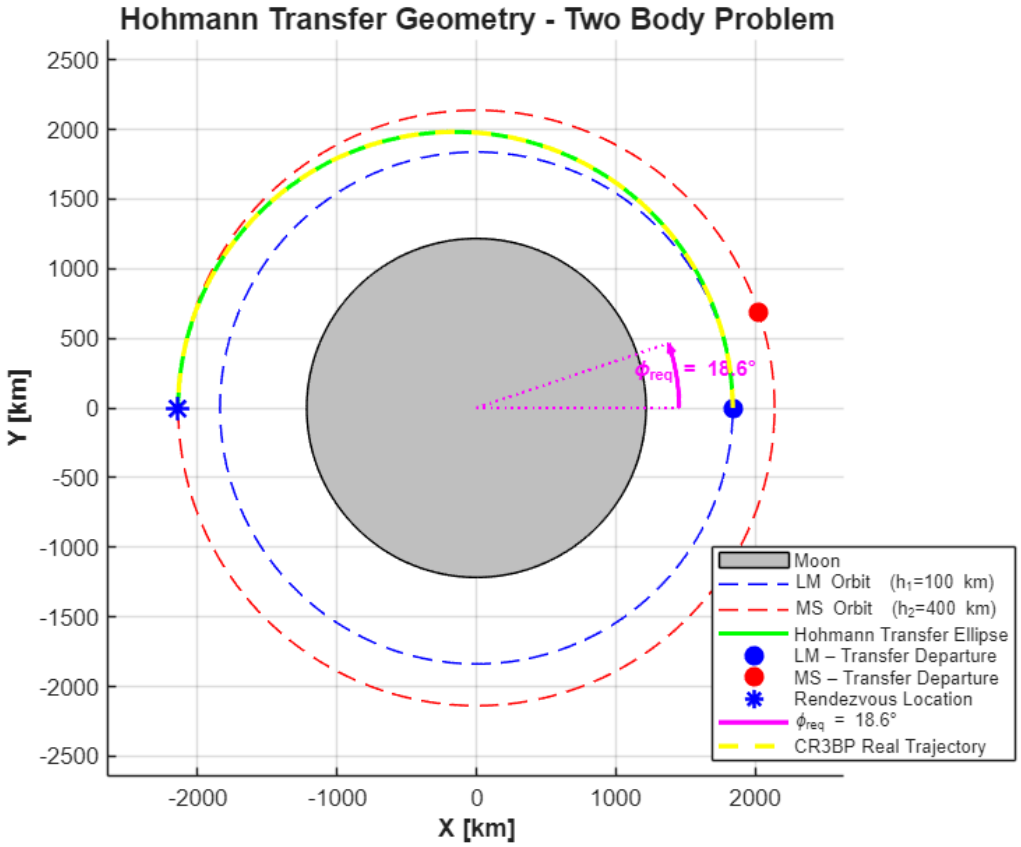


Figure 25: Transfer trajectory in the CR3BP (yellow dashed) overlaid on the two-body Hohmann geometry (green). The two curves are nearly indistinguishable at this scale, confirming the negligible effect of the Earth's tidal perturbation for this short-range ($\Delta h = 300$ km) lunar transfer.

6.1.3 Discussion

The absolute error of the Jacobi constant $|C(t) - C(0)|$ is plotted in Figure 24. The error remains below 5×10^{-11} throughout the integration, confirming that the equations of motion are implemented correctly

and that the `ode45` tolerances are appropriate. The oscillatory pattern visible in the plot is due to numerical noise introduced by the adaptive step-size algorithm of the integrator, not a physical variation of C.

The CR3BP results comparison with the two-body Hohmann results from Section 2 is developed in Table 22. The ΔV values are identical to four decimal places. This is physically expected: the Earth's tidal acceleration at LLO altitude is $a_{3B} \sim 10^{-5}$ m/s^2, which over the transfer time $\Delta t \approx 4000$ s produces a velocity correction of order

$$\delta V \sim a_{3B}\,\Delta t \approx 10^{-5} \times 4000 = 4 \times 10^{-2} \text{ m/s} = 4 \times 10^{-5} \text{ km/s}, \tag{115}$$

below the fourth decimal place. The CR3BP trajectory costs the same propellant as the Hohmann transfer but follows a slightly different path (Figure 25), resulting in a ~ 16 s longer time of flight.

The shooting method is essential to correct the radial altitude error that the theoretical two-body Hohmann solution produces when applied in the perturbed three-body environment. Without correction, Earth's gravity introduces a small but critical radial mismatch at the target: propagating the nominal Hohmann parameters yields an achieved radius of 2137.325 km, missing the target MS orbit (2137.400 km) by approximately 74.77 m. The `fsolve` iteration dynamically adjusts the initial velocity and time of flight to perfectly eliminate this mismatch, ensuring an accurate rendezvous (within numerical tolerances) while requiring only a negligible adjustment to the overall propellant cost.

Role of the Lagrange points The five Lagrange points L_1–L_5 of the Earth-Moon CR3BP are equilibrium positions in the rotating frame where the net effective force vanishes. For $\mu = 0.01215$:

- L_1, between Earth and Moon ($\approx 58\,000$ km from the Moon), is the gateway for low-energy transfers from Earth. Trajectories passing near L_1 can reach LLO with significantly less ΔV than a direct Hohmann, at the cost of weeks to months of transfer time.
- L_2, beyond the Moon on the Earth–Moon line ($\approx 65\,000$ km), is used for distant retrograde orbits and the Gateway lunar station.
- L_4 and L_5 are stable equilateral triangle points; they play no role in transfer missions.

In our scenario the transfer covers only 300 km of altitude change, entirely within the Moon's sphere of influence ($r_{SOI} \approx 66\,000$ km). The Lagrange points are ~ 200 times farther away, so their influence on this short-range maneuver is negligible. This is confirmed by the numerical result: the CR3BP and Hohmann solutions are identical to four significant figures.

6.2 Inverse Kinematics and Trajectory Planning

Continuing from Section 5, solution of the inverse kinematic problem for the manipulator and joint-space trajectory planning from a rest configuration to an arbitrary joint configuration θ^* within the reachable workspace.

6.2.1 Methodology

The inverse kinematics problem consists of finding the joint configuration $\theta^* \in \mathbb{R}^5$ such that the end-effector reaches a prescribed target position $p_{target} \in \mathbb{R}^3$ expressed in the base frame.

Iterative method A Jacobian based iterative method is adopted, as described in [11]. The iterative update law reads:

$$\theta_{k+1} = \theta_k + K \cdot J^* \cdot \Delta x_k \tag{116}$$

where $\Delta x_k = p_{target} - p_{EE}(\theta_k)$ is the Cartesian position error at iteration k, K is a scalar step gain, and J^* is the regularised pseudo-inverse of the Jacobian.

The geometric Jacobian $J \in \mathbb{R}^{6\times5}$ is constructed following the standard formulation for revolute joints [11]. Since the manipulator consists exclusively of revolute joints, each column takes the form:

$$J_i = \begin{bmatrix} z_{i-1} \times (p_{EE} - p_{i-1}) \\ z_{i-1} \end{bmatrix} \tag{117}$$

where z_{i-1} is the rotations axis of joint i expressed in the inertial frame, and p_{i-1} is the origin of the link's $\mathcal{L}_{i-1}$ frame. Note that linear part already accounts for the distance to the end-effector tip directly, without requiring any additional shift transformation.

During the first implementation, the algorithm attempted to control both position and orientation simultaneously, using the full 6×5 Jacobian and a six-dimensional error vector:

$$\Delta x = \begin{bmatrix} p_{target} - p_{EE} \\ \delta\omega \end{bmatrix} \in \mathbb{R}^6 \tag{118}$$

where the orientation error $\delta\omega$ is computed via the Lie Algebra's *vee* operator applied to the skew-symmetric part of the rotation error matrix [12]:

$$R_{\text{err}} = R_{\text{target}} R_{\text{curr}}^T, \qquad \delta\omega = \begin{bmatrix} R_{\text{err}}(3,2) - R_{\text{err}}(2,3) \\ R_{\text{err}}(1,3) - R_{\text{err}}(3,1) \\ R_{\text{err}}(2,1) - R_{\text{err}}(1,2) \end{bmatrix} \tag{119}$$

This initial formulation resulted in a persistent residual error of approximately 5 m. As shown by the Singular Value Decomposition of the full 6×5 Jacobian:

$$\sigma = [4.0796 \quad 2.1778 \quad 2.1464 \quad 0.2885 \quad 0.0000], \tag{120}$$

one singular value is identically zero, indicating that the 6×5 Jacobian has structural rank 4 rather than 5 for the adopted kinematic configuration. This rank deficiency implies that there exist one direction in the joint space that produces no end-effector motion, reducing the effective controllable degrees of freedom in the full task space. As a consequence, the full 6×5 Jacobian cannot independently control all six components of the end-effector twist simultaneously. To overcome structural rank deficiencies encountered with the full Jacobian, the IK problem is reformulated using only the linear part of the Jacobian $J_{pos} \in \mathbb{R}^{3\times5}$, which maps joint velocities to end-effector linear velocity only. This reduces the problem to 3 equations in 5 unknowns, yielding a redundant system with 2 null-space degrees of freedom. This position-only formulation is also consistent with the berthing mission requirement, where accurate end-effector positioning at the docking port is the primary objective, while the final orientation can be adjusted independently by the last joint.

Damped Least Squares It is important to underline that the inversion of the Jacobian can represent a serious inconvenience not only at a singularity but also in the neighbourhood of a singularity. For instance,

for the Jacobian inverse it is well known that its computation requires the computation of the determinant; in the neighbourhood of a singularity, the determinant takes on a relatively small values which can cause large joint velocities. An alternative solution overcoming the problem of inverting Differential Kinematics in the neighbourhood of a singularity is provided by the so-called damped least-squares (DLS) inverse:

$$J^* = J^T(JJ^T + k^2 I)^{-1} \tag{121}$$

where k is a damping factor that renders the inversion better conditioned from a numerical viewpoint. [11]

Target Selection A critical aspect of the IK problem is ensuring that the target lies within the reachable workspace of the manipulator. The target position $p_{target} = [4, 0, 1]^T$ m was selected after verifying its membership in the reachable workspace. The initial joint configuration is set to $\theta_0 = \left[0, 0, 0, \frac{\pi}{2}, 0\right]^T$ rad.

Trajectory Planning Starting from θ_0 and targeting the IK solution θ^*, a rest-to-rest joint-space trajectory is planned using cubic polynomial interpolation over a motion time $T = 1800$ s, representative for a typical berthing operation. For each joint i, the trajectory is [11]

$$\theta_i(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 \tag{122}$$

with boundary conditions $\theta_i(0) = \theta_{0,i}$, $\theta_i(T) = \theta_i^*$, $\dot{\theta}_i(0) = \dot{\theta}_i(T) = 0$, yielding:

$$a_0 = \theta_{0,i}, \quad a_1 = 0, \quad a_2 = \frac{3\Delta\theta_i}{T^2}, \quad a_3 = \frac{-2\Delta\theta_i}{T^3} \tag{123}$$

6.2.2 Results

Inverse Kinematics Solution The iterative method implemented converges to the required joint angular displacement needed to reach the target position $p_{target} = [4, 0, 1]^T$ m:

$$\theta^* = [0, 0, 16.2625, -0.0018, 79.5438]^T \text{ deg} \tag{124}$$

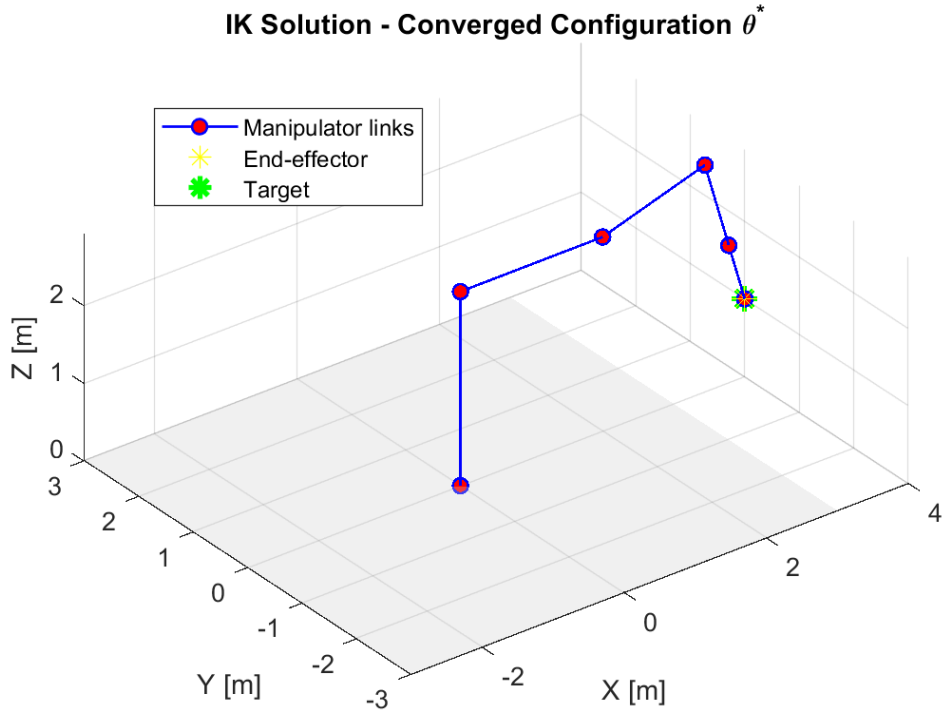


Figure 26: Manipulator configuration needed to reach the target End-Effector position of $p_{target} = [4, 0, 1]^T$ m)

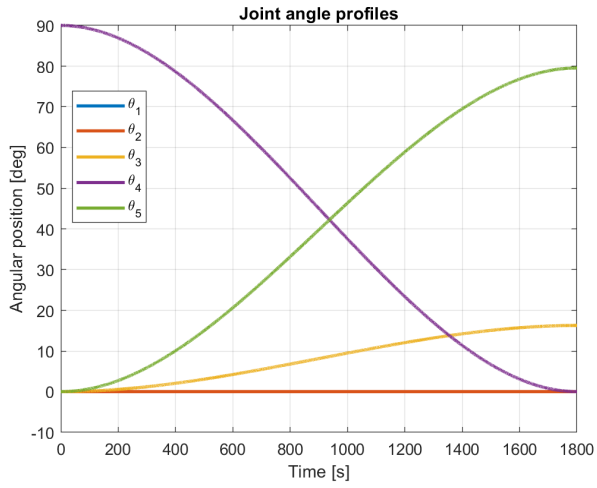


Figure 27: Joint angle profile: the value of the joints angle varies in time during the motion of the arm.

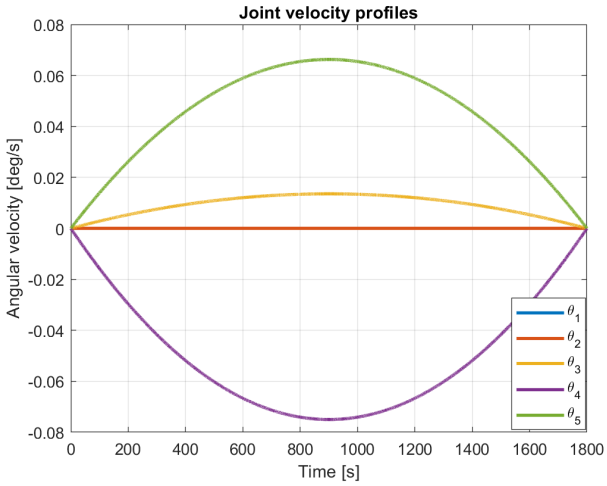


Figure 28: Joint velocity profile: the value of the joints velocities varies in time during the motion of the arm, but return to zero, confirm the respect of the rest-to-rest condition.

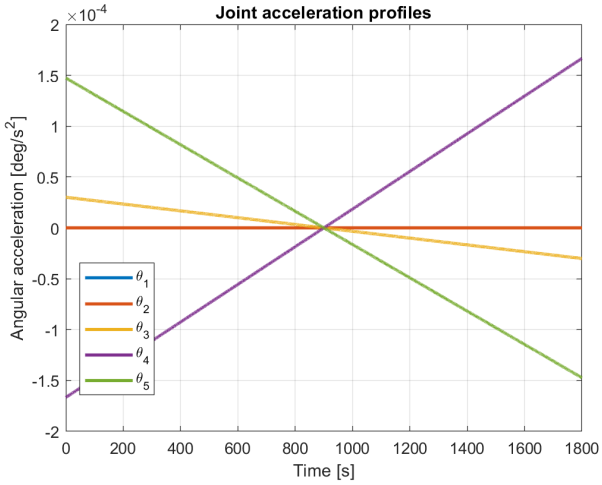


Figure 29: Joint acceleration profile: the value of the joints acceleration evolves in time during the motion of the arm with a linear behaviour.

6.2.3 Discussion

It is possible to see from equation (124) and Figure 26 that, while the base joint (θ_1) and the second joint (θ_2) remain at 0° because the target lies on the xz-plane, reaching the extended distance along the X-axis forces a major reconfiguration of the subsequent links: the iterative algorithm commands the third joint to rotate upward ($\theta_3^* \approx 16.26°$) in order to elevate the end effector. Then, the end-effector wrist is rotated ($\theta_5^* \approx 79.54°$) to reach the target position. The algorithm achieves this with a negligible norm error of 8.46×10^{-5}, verifying both workspace membership and solver accuracy.

When changing the target position to $p'_{target} = [4, 0, 3]^T$ m, the norm error is 0.75387 that is significantly higher than in the previous case. In fact, from [11] we can see that the convergence of the Jacobian-based IK algorithm is sensitive to the choice of target position, even when the target lies within the reachable

workspace. This behaviour is attributed to the local nature of the iterative method: the algorithm follows a gradient-descent path in joint space that depends on the Jacobian evaluated at each intermediate configuration. Certain targets require the manipulator to traverse near-singular regions along the implicit path from θ_0 to θ^*, causing the condition number of J_{pos} to grow and the convergence to deteriorate. This is a known limitation of first-order Jacobian methods, which only guarantee local convergence and are sensitive to the initial configuration and the geometry of the workspace.

In the trajectory planning phase, we verified that the resulting joint velocity profiles are parabolic with zero values at both endpoints, confirming the rest-to-rest condition. The acceleration profiles are linear in time, with non-zero values at $t = 0$ and $t = T$, which is a known limitation of cubic interpolation. Continuity of acceleration at the endpoints could be recovered by adopting a quintic polynomial at the cost of two additional boundary conditions.

In all of the joint profiles above (Figure 27, 28, 29), the value of θ_1 does not appear because its value is fixed and therefore does not change in time. Moreover, starting from the joint angle profile, we can see that the quantities behaviour shift from a cubic behaviour, to a parabolic behaviour to a linear behaviour as expected, these quantities being derivatives of one another. It is important to notice that at $t = 900$ s the velocity profile for each joint reaches the maximum absolute value and the acceleration profiles of all joints meet at 0 deg/s: the joint motors have reached the peek velocoty and from now on they will start to slowly decelerate in order to reach the final position at rest (zero velocity). Moreover, the acceleration profile meet all at the same point because they're modelled with the same cubic polynomial interpolation: therefore, their movement is synchronized.

6.3 Lunar Lander Survey

Since the beginning of the Space Age, landing on the Moon was one of the principal objectives of the space agencies all around the world. The great unknowns were to assess if the lunar surface was solid enough to sustain the lander weight and if the lander could lift off again from the surface in order to retrieve humans or scientific samples. Different solutions were adopted and many landers succeeded in getting safely to the lunar surface.

Apollo Lunar Module (NASA) Nicknamed "Eagle", it was the first lander to allow humans to step on the Moon, on July 20th 1969, during the NASA Apollo 11 mission. Being part of a human mission, it was of utmost importance to bring back safely the astronaut. It was composed of two main stages, one for the descent and the other for the ascent. The descent stage had an initially stowed 4 legged landing gear, a throttleable, gimballed engine which provides from 1,050 to 9,710 pounds of thrust. It is now on the lunar surface, as it served as launch pad for the ascent stage. The ascent stage was the only habitable space during the surface mission. It was equipped with all the life support systems, the propellant tank for its dedicated ascent engine, the reaction Control System for attitude correction and an Inertial Measurement Unit, needed to obtain the ascent stage angular rate. During the ascent phase, it was jettisoned from the descent module and inserted the Apollo capsule into Lunar orbit to rendezvous with the Command and Service Module (CSM).
Its major criticality was the fact that the ascent engine was a non-throttleable, hypergolic pressure-fed engine, where the propellants ignited at contact simplifying the engine architecture. This meant that all the electrical and plumbing connection had to be severed completely and at the right instant in order to allow the engine to lift the ascent stage weight. If some appendages were to remain attached, considering the case of the engine being able to provide enough thrust to rip out the constraint, this delay would have

wasted part of the propulsive power needed to reach the correct orbit with the right velocity, from which the stage would have performed a series of burns in order to rendezvous with the CSM, much alike the operations modelled in our scenario [3].

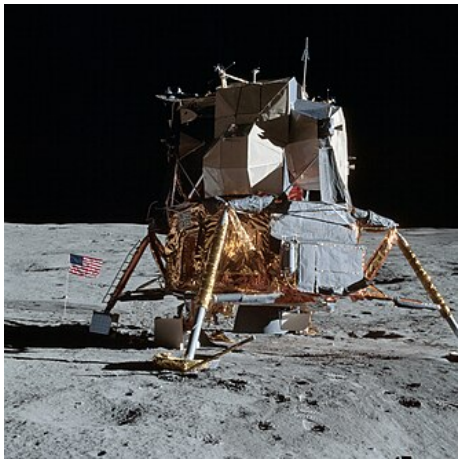


Figure 30: The Apollo 14 Lunar Module on the lunar surface [3].

Chandrayaan-3 (ISRO) The Indian Space Agency managed to land this spacecraft at the lunar South Pole on August 23rd 2023. The Integrated Module is composed of a Lander Module (LM), a Propulsion Module (PM) and a Rover. The lander soft landed on the lunar surface and deployed the rover that performed in-situ chemical analysis of the lunar surface. Even though the lander engine wasn't specifically designed for ascent, it managed to reignite and lift off to relocate at the landing site. [4]



Figure 31: The Chandrayaan-3 Lander Module portrayed by the mission rover deployed on the surface of the Moon [4].

Blue Ghost - Mission 1 (Firefly Aerospace) On March 2nd 2025 this commercial mission landed in the Mare Crisium area on the Moon. The mission goal was to deliver a series of scientific and research demonstrators such as the Lunar GNSS Receiver Experiment (LuGRE) developed with the participation of the Italian Space Agency (ASI). The importance of this mission lays in the fact that it was the longest commercial mission on the lunar surface. In the descent phase, the engine is shutdown and the spacecraft

is controlled by the reaction control system (RCS) thrusters, landing with a vertical velocity of 1 m/s: in the perspective of a hypothetical ascent from the surface (for example, in the context of a sample return mission), this engine needs to be designed as a reusable engine with the ability to be reignited. It is also necessary to take into account the propellant storage and the fact that the payload has to withstand the vertical G-force experienced during the sudden ascent burn [8].

Starship HLS (SpaceX) A very debated project, it promises to land humans on the Moon in the context of the NASA Artemis IV mission. The lander is a 52 m tall, single stage spacecraft with multiple reusable engines. After being used as a second stage during the Earth ascent and for the Translunar Injection (TLI) maneuver , the lander engine will need to reignite again for the descent on the Moon and finally for the ascent phase. Unlike the Apollo Lunar Module, it will not leave any structure on the surface, carrying all its mass back to Earth. Its dimensions and the necessity of engine reusability pose a serious criticality in the mission: the engine may be damaged during the flight and need to be optimized for different ambient condition (atmospheric flight, void): a solution to this problem may be using different engines used in different phases of the mission. Moreover, the ullage volume of 163 m^3 during the Moon ascent phase will be mostly empty, causing the Center of Gravity of the spacecraft to vary significantly, shifting to the bottom side of the lander. This could hinder the spacecraft dynamics, and translate into a need for an oversized AOCS (Attitude and Orbit Control System) to orient and control the attitude of the spacecraft. Finally, the same propellant used by the engine (LOX-LCH4) pose a serious problem, being not easily storable and needing in-orbit refuelling: rendezvous and docking operations are then to be planned, and performing such operations with a big and not at all agile spacecraft, is no easy task [6].

Blue Moon (Blue Origin) Blue Origin promises to bring humans and cargo to the Moon in an affordable and reusable lander, allowing a stable presence on the surface in the near future. As of today, the Mark 1 model is undergoing tests in NASA's facilities, as part of a future NASA Artemis mission. The actual crewed spacecraft is the Mark 2 model and it will be 16 meters tall, with a mass of 16 ton and it will be propelled by a LOX-LH2 reusable engine. The same criticalities of Starship HLS could be addressed for Blue Moon, with the addition of the need to prevent the boil-off of the cryogenic propellants. Its maneuverability seems somehow better, being smaller. It is planned to be transferred to and from the lunar orbit by the Cislunar Transporter by Lockheed Martin: rendezvous and docking operations are needed, as well as in orbit refuelling, accounting for the propellant boil-off rate during all the mission phases [7].

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